

Remainder control, physical readouts and homogeneous observables: a Round28 research addendum

Research workbench draft for author review

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Abstract

This additive manuscript records ten admitted investigations in the Round28 gauge-theory workbench. The homogeneous branch reconstructs a complete original remainder and controls one correction; later investigations establish a locally normal physical representation, a nonzero original Wilson fluctuation and a conditional quantitative variance floor. A physical form-domain argument then bounds that fluctuation's first energy moment, certifies mass in a finite spectral interval and gives a lower imaginary-time correlation envelope. Separate canonical readout results retain complete prediction errors, while a larger finite physical graph receives exact operator data and all-time heat-error certificates. These models remain distinct. Scoped derivations, reproducible calculations, negative controls and an attributed source survey support each conclusion. No nontrivial four-dimensional continuum Yang–Mills construction or continuum physical mass gap is claimed.

Draft and attribution. Human authorship and affiliations remain to be supplied. No arXiv submission is claimed. Historical names denote contemporary methodological lenses, not participation or endorsement. The two-star marker ** denotes a derivation or implementation within this workbench; scientific priority is unverified.

Final scoped record: 10 of 10 requested investigations have included advisor gates. Task completion is distinct from theorem completion.

1 Inherited obligations and adaptive research design

Round27 established an actual one-step correction for the selected cubic-source sector $G + A_3$, with a narrower-coupling contraction of its selected mixing family and explicit support-weight loss. Its extra original remainder E remained conditional. The first Round28 question is therefore whether that omitted interaction can be completely reconstructed and controlled. A positive bound for an extracted reference operator must remain separate from a gap of the full Hamiltonian with its remaining mixing.

The authorized continuation has five goal pairs with two reviewed investigations per pair. The first three inherited branches concern homogeneous remainder control, cancellation-aware canonical readouts and larger physical graphs. Their six reviews led to the homogeneous representation and Wilson-nonvacuity pair; that review then led to the quantitative observable goal. Every second contract is selected after its first review, and only gated results enter this manuscript. Repairs, literature searches, replays and refinements within a frozen contract are not additional research loops.

The advisor freezes each model, observable, domain, parameter interval and desired conclusion. Separate forward and reverse researchers work before exchanging current results; a separately authored skeptic challenges the proof and exact checks. Shared inherited premises mean these are

independent derivations and implementations, not independent physical observations or external peer review. Accepted gates control which chapters enter the compiled manuscript.

Physical units remain explicit. Fixed positive α , $\hbar$ and E_* cannot be recalibrated to hide a failed matching estimate. In the homogeneous normalized Hamiltonians, physical energy is obtained using $\delta = \alpha/8$, with original spacing fixed. Support weights and filter durations are proof parameters. Parameters in the canonical summable family are not automatically a regulator trajectory or a homogeneous-model identification.

2 A bounded, attributed source survey

The initial Round28 combined ledger records **30 formal readings at 38 distinct recorded URLs**. A record is one expert-source entry or skeptic external-source entry, not one completely read book or independent study. Several lenses revisit the same paper; one entry can include abstract, PDF and publisher URLs. Three separate screening/reopening entries bring the URL union to 41. One additional unread lead would bring it to 42, but is excluded from the reviewed-source count. These distinctions are preserved in `research/round28/experts/sources.json`. The later source-only addition and its combined counts are recorded separately in Appendix B.

Lens	Records	URLs	Reading character
Newton	3	3	Two historical manuscripts and deeper selected stability-paper passages.
Tesla	5	5	Autobiography, patent, forum claims and two 2026 cancellation papers.
Jung/Pauli	9	9	Historical writings at different depths, two discussions and two inference papers.
Penrose	5	11	Stability, gauge reduction, variational benchmarks, exact integration and a channel extension.
Feynman	5	10	Stability, local transformations, exact algebra and two bounded current-claim inspections.
Skeptic	3	3	Official problem/status records and an older-paper bibliographic check.

Table 1: Per-lens URL counts overlap and must not be summed into a unique-source count. The survey itself executes zero physics loops.

The categories comprise eight historical-primary records, sixteen technical-primary records, one patent, three forum records and two official problem references. Primary provenance does not establish reliability, applicability or scientific priority. Newton explicitly deepened the historical ALCH00081 reading and selected sections of an inherited stability paper. Other new ledger entries are new recorded readings by that lens, not claims of first project discovery. This manuscript attributes the recorded findings; it does not claim a fresh audit of every external paper or full proof.

2.1 Historical and unconventional material: questions, not premises

Newton’s selected folios of *Of Natures obvious laws* motivate asking which terms or reservoirs have been omitted from a proposed mechanism. His [October 1666 fluxions tract](#), Proposition 7 and selected reconstruction passages, motivates testing whether an inverse returns the originating relation. Missing rendered formulas prevent a complete transcription or derivation audit. Neither source validates an alchemical mechanism or supplies a modern gauge-theory term.

Tesla’s *My Inventions*, section VI was inspected in a transcription at selected retrospective mystical and mechanistic passages. It is neither the original magazine scan nor a controlled observation. [US685957A](#), around figures 1, 3 and 4 and the claims, specifies incident radiation, storage, receiver control and output conditions. Its proposed apparatus does not establish unrestricted loading or

instantaneous transmission. The associated forum exchange is a record of claims to test against the patent, not corroborating physical evidence.

Jung’s *Collected Papers on Analytical Psychology* was read in the verified 1920 second-edition reprint: selected occult-phenomena, number-dream and rumour passages are three writings at one URL, not three books. Publisher opening excerpts of the *I Ching* foreword and “Psychology and Spiritualism” do not amount to reading the full essays. The operational question is whether a fixed interpretation separates a prediction from post hoc selection, remembered matches and ignored misses. Two purposively sampled discussions do not establish prevalence, authenticate every circulating quotation or verify paranormal effects.

There is **no newly read government-primary source** in these Round28 ledgers and no newly inspected original Pauli letter. Earlier FBI, DOE and CIA material remains in Round27 at its original reading depth and is not counted again. Clay is an official mathematical-problem source, not a government program. Historical or mystical interpretations add no Hamiltonian coefficient, measured frequency, physical clock or continuum map.

2.2 Technical comparisons and remaining applicability tests

Three lenses revisit [Del Vecchio–Fröhlich–Pizzo, arXiv:2108.13907](#), with selected induction, domain, assumption and theorem passages recorded separately. The 2021 preprint and 2023 journal publication motivate an applicability audit. An existential stability threshold cannot be cited as an evaluated numerical threshold or a completed workbench induction. Selected local-transformation passages in [Bravyi–DiVincenzo–Loss, arXiv:1105.0675](#) and exact-algebra passages in [Ying–Li–Po, arXiv:2509.12621](#) reinforce full transformed-operator accounting. Spin and qubit fixtures cannot replace the full $SU(2)$ support and domain arguments.

Tesla’s selected [noise-canceling observable](#) and [spacetime mitigation](#) papers motivate retaining model bias, joint omitted events, normalization and sampling resources. They imply no automatic cancellation of this workbench’s error disks. Jung’s [relative-fit inference comparison](#) distinguishes inference under misspecification from a true-model statement. The [Hamiltonian observation-geometry comparison](#) makes sampled data a parameter pseudometric unless injectivity is proved. Deterministic interval bounds have no statistical coverage interpretation without a sampling model.

Penrose’s selected [gauge-fixed Hamiltonian](#) and [variational ground-state](#) comparisons require the full induced generator, Gram matrix and omitted channels. A variational energy is not a certified gap. [Dahlqvist’s exact integration work](#) motivates exact quotient calculations, with its surface, group and limit assumptions retained. A [quantum-channel spin-network extension](#) changes the dynamics and is not silently imported into a closed Hamiltonian model.

The supplemental Feynman review inspects selected assumptions and theorem passages of the [Faizal–Shabir four-dimensional construction claim](#); it does not audit the entire 593-page compilation. Questions about comparison direction, quantitative defect margins, time factors and tuning-map rank require checking the other supporting sections before a conclusion. The panel neither validates nor refutes the whole construction. [Nair’s lower-dimensional proof outline](#) was inspected at selected framework, equations and regularization passages. Its closed domains, regulator limits and vacuum/energy centering remain applicability obligations; it is not a four-dimensional matching theorem. The formalization lead [2603.15770](#) was unread by that lens in this pass and excluded from its reviewed count.

Version discrepancies remain explicit. Two experimental HTML headers display dates different from their version histories; no later revision is inferred. Dahlqvist’s landing abstract and version-2 theorem use different genus thresholds, so the specified theorem/version controls the theorem-scope claim. The two different math-ph identifiers in screening and bibliographic records are not merged.

Root metadata checks are separate from full-proof audits. The ledger retains exact source pointers and reading limits rather than hiding these discrepancies.

How the survey changes the tests. For the homogeneous branch, deleting a quadratic class, overlapping ordered word, factorial coefficient, scalar correction or transported remainder must fail reconstruction. For canonical readouts, retain higher odd terms, joint/adversarial error, phase and clock uncertainty and denominators containing zero; a cancellation coefficient containing unknown q must be candidate-indexed or carry its estimation error. For graph enlargement, require exact Gram null relations, every reached and omitted channel, boundary return and an added-region observable. These are source-motivated controls for frozen goals, not new physical premises or extra loops.

The [official status](#) and [Jaffe–Witten problem statement](#) remain the target references. Nontrivial four-dimensional construction, the required axioms, compact simple groups, a positive physical mass gap and scale matching remain distinct obligations. Source counts, network edges and loop completion supply no defensible percentage of that theorem.

3 AG2: the complete original remainder and its reference^{**}

The first admitted investigation reconstructs the original homogeneous remainder that was conditional in AG1 and transports every class through the actual AG1 correction. On $0 \leq M \leq 10^{-3}$ the resulting indexed weight-two interaction obeys $\|K\|_2 < 7/1000$; on $0 \leq M \leq 10^{-4}$ it obeys $\|K\|_2 < 7/100000$. The centered reference alone has normalized gap greater than 0.995. The full corrected Hamiltonian retains a mixing operator, so this is not its gap theorem.

3.1 Domain-preserving transformation and a four-part inventory

Keep the homogeneous complete 24-link coarse factors and all 21 omitted faces on each retained complete star $Z_b = b + \{0, e_1, e_2, e_3\}$. The onsite reference is $H_0 = \sum_x h_x$, with $h_x \geq I - P_x$ and product vacuum Ω . Write

$$\begin{aligned} v_b &= \phi_b \Omega_b, \quad u_b = (H_{0,Z_b}|_{Q_b})^{-1} v_b, \quad L_b = |u_b\rangle\langle\Omega_b| - \text{adjoint}, \\ A_{1,b} &= |v_b\rangle\langle\Omega_b| + \text{adjoint}, \quad D_b = Q_b \phi_b Q_b, \quad X = \sum_b L_b, \quad G = H_0 + D. \end{aligned} \quad (1)$$

Here $\Phi = \sum_b \phi_b$, $A_1 = \sum_b A_{1,b}$, $D = \sum_b D_b$, $M = 7|\tau|$ and $\|\phi_b\| \leq M$. Exact Haar orthogonality gives $\|v_b\|^2 = M^2/84$. Rational upper bounds are $\|L_b\| \leq s$ and $\|A_{1,b}\| \leq \sigma$, where $s = \sigma = 4M/35$. They are not evaluated norm equalities.

The actual vectors u_b, Ω_b lie in the local reference domain, and $[L_b, H_0] = -A_{1,b}$ has a bounded extension. Strong commutation with exterior energy and graph-norm control justify $e^{\pm X} D(H_0) = D(H_0)$ in finite cuboids. Integrating the first bounded commutator before expanding the remaining bounded terms gives

$$\overline{H} = e^X (H_0 + \Phi) e^{-X} = G + R_{O1}, \quad R_{O1} = \sum_{n \geq 1} \left(\frac{\text{ad}_X^n \Phi}{n!} - \frac{\text{ad}_X^n A_1}{(n+1)!} \right). \quad (2)$$

This does not postulate an operator-norm BCH series for unbounded H_0 .

Pair the two terms at each ordered anchor history, retaining repetitions and the complete star union. At a same-anchor depth-two history define

$$C_b = \frac{1}{2} \text{ad}_{L_b}^2 \phi_b - \frac{1}{6} \text{ad}_{L_b}^2 A_{1,b}, \quad A_{3,b} = P_b C_b Q_b + Q_b C_b P_b. \quad (3)$$

This $A_3 = \sum_b A_{3,b}$ is precisely the earlier selected cubic source. It is distinct from A_1 , the total cubic coefficient and the full remainder. The complete missing interaction is

$$\begin{aligned} E &= R_{O1} - A_3 = E_2 + E_{3,\text{same}} + E_{3,\text{other}} + E_{\geq 4}, \\ E_2 &= [X, D] + \tfrac{1}{2}[X, A_1], \quad E_{3,\text{same}} = \sum_b (C_b - A_{3,b}). \end{aligned} \quad (4)$$

$E_{3,\text{other}}$ contains every other ordered depth-two history; $E_{\geq 4}$ contains all depths at least three with their original factorials. The first three support maxima are seven, four and ten sites; the tail has arbitrary finite connected supports. These labels specify lowest algebraic degrees 2, 3, 3, 4, not nonvanishing of every class.

The same-anchor subtraction is an orthogonal block compression: $C_b - A_{3,b} = P_b C_b P_b + Q_b C_b Q_b$, of norm at most $\|C_b\|$. Its centered split is $c_b I + Q_b(C_b - c_b I)Q_b$; using $c_b I + Q_b C_b Q_b$ would count $c_b Q_b$ twice. In each fixed nonempty cuboid the actual vacuum expectation has a nonzero quadratic term $\langle \Omega, E\Omega \rangle = -a_\Lambda \tau^2 + O_\Lambda(|\tau|^3)$ with $a_\Lambda > 0$. Thus E cannot be a quartic-only operator-norm error in that volume. This neither computes its quadratic mixing nor proves a uniform lower coefficient.

3.2 Complete low-order geometry and the all-order tail

For the declared indexed norm $\|B\|_a = \sup_x \sum_{Y \ni x} a^{|Y|} \|B_Y\|$, let $Q_n(a)$ count ordered relative histories with the size of their complete union as root multiplicity. The exact first polynomials are

$$\begin{aligned} Q_0(a) &= 4a^4, \quad Q_1(a) = 4a^4 + 84a^7, \\ Q_2(a) &= 4a^4 + 252a^7 + 216a^9 + 1920a^{10}. \end{aligned} \quad (5)$$

An incoming anchor meeting a union Y is exactly $y - z$ for $y \in Y$, $z \in Z_0$. Enumerating those finite sets gives depth-two history counts $(1, 36, 24, 192)$ at sizes $(4, 7, 9, 10)$. Finite boundaries delete histories; no interactions outside the original boundary are introduced.

Finite enumeration is not the infinite-tail proof. At depth n every connected union has at most $4 + 3n$ sites. Counting both root locations, repeated anchors and incoming overlaps gives

$$N_{n+1}(a) \leq 8sa^3(8 + 3n)N_n(a), \quad N_n(a) \leq 4a^4b(24sa^3)^n(8/3)_n \quad (6)$$

for a seed-norm bound b . With $x = 24sa^3$, the factorwise comparison $(8/3)_n/n! \leq (n+1)(n+2)/2$ supplies the generating function $(1-x)^{-3}$. A complete positive majorant is

$$\begin{aligned} e_2(a) &= 2s(M + \sigma/2)Q_1(a), \quad u_3 = 4s^2(M/2 + \sigma/6), \\ e_{3,\text{same}}(a) &= 4a^4u_3, \quad e_{3,\text{other}}(a) = [Q_2(a) - 4a^4]u_3, \\ e_{\geq 4}(a) &= 4a^4(M + \sigma/4)[(1-x)^{-3} - 1 - 3x - 6x^2], \\ \|E\|_a &\leq e(a) := e_2(a) + e_{3,\text{same}}(a) + e_{3,\text{other}}(a) + e_{\geq 4}(a). \end{aligned} \quad (7)$$

At $a = \rho = 9/4$, $x = (2187/70)M \leq 2187/70000 < 1$ on the main interval. Nonnegative power-series coefficients and monotonicity justify continuous interval bounds from exact rational endpoint evaluations. The same-anchor compression is counted once; its selected mixing norm is not added to an already complete cubic budget.

3.3 Transporting every original term

Let S be the admitted AG1 generator, distinct from X , with $[S, G] = -A_3 + R_{A_3}$. The complete corrected operator is

$$\begin{aligned} e^S \bar{H} e^{-S} &= G + K, \quad K = R_{A_3} + N + e^{\text{ads}} E, \\ N &= \sum_{n \geq 1} \frac{\text{ad}_S^n(nA_3 + R_{A_3})}{(n+1)!}. \end{aligned} \quad (8)$$

Graph-domain preservation applies to the actual S ; finite-volume E is bounded and need not separately regularize arbitrary domain vectors. With $r_{\text{bd}} = M^3/567$, $A_{3,\rho}^{\text{bd}} = 4\rho^4 r_{\text{bd}}$ and $F = (1 - 72\rho^3 M)^{-3}$, take

$$\theta = 18A_{3,\rho}^{\text{bd}}F < 1, \quad n_2 = \frac{\theta}{1-\theta}A_{3,\rho}^{\text{bd}}(1+F/2), \quad \|K\|_2 \leq k(M) := r_2 + n_2 + \frac{e(\rho)}{1-\theta}. \quad (9)$$

Here $r_2 = 64r_{\text{bd}}(1 - 576M)^{-3}$ is valid on the main interval. Only on the inherited narrow interval may it be replaced by $64r_{\text{bd}}[4/9 + (1 - 576M)^{-3} - 1]$. The geometric transport factor follows by allocating the ρ -to-two weight loss over every nested commutator. The entire margin is spent; the resulting source does not regain a stronger norm by renaming its weight. The reviewed conservative bounds are $k(M) < 0.007$ on the main interval and $k(M) < 0.00007$ on the narrow interval.

3.4 The centered reference bound and its exact limitation

For every generated self-adjoint K_Y , including transported E , retain its full support and split

$$\begin{aligned} c_Y &= \langle \Omega_Y, K_Y \Omega_Y \rangle, \quad B_Y = P_Y K_Y Q_Y + Q_Y K_Y P_Y, \\ Z_Y &= Q_Y (K_Y - c_Y I) Q_Y, \quad K_Y = c_Y I + B_Y + Z_Y. \end{aligned} \quad (10)$$

The separate norm costs are one, one and two times $\|K_Y\|$. Every assigned support has at least four sites, including scalar terms. Double counting over sites and $1/(|Y|2^{|Y|}) \leq 1/64$ give

$$\frac{|C_\Lambda|}{|\Lambda|} \leq \frac{k(M)}{64}, \quad \|B\|_2 \leq k(M), \quad \|Z\|_2 \leq 2k(M), \quad C_\Lambda = \sum_Y c_Y. \quad (11)$$

Because Z_Y annihilates its local vacuum and $Q_Y \leq \sum_{x \in Y} (I - P_x)$,

$$|\langle \psi, (D + Z)\psi \rangle| \leq \left(4M + \frac{k(M)}{8}\right) \langle \psi, H_0 \psi \rangle. \quad (12)$$

The factor $k/8 = 2 \cdot 2^{-4}k$ retains the four-site support minimum. Thus the centered reference, self-adjoint on $D(H_0)$ in each finite cuboid, satisfies

$$G_{\text{ref}} = H_0 + D + Z \geq \left(1 - 4M - \frac{k(M)}{8}\right) H_0 \geq \left(1 - 4M - \frac{k(M)}{8}\right) (I - P_\Omega). \quad (13)$$

The main interval therefore has a reference-only normalized gap greater than 0.995; physical energies multiply by $\delta = \alpha/8$. Adding $C_\Lambda I$ shifts this reference ground energy. The actual corrected Hamiltonian is instead $C_\Lambda I + G_{\text{ref}} + B$, with B still present. No gap for it, contraction of its complete mixing, next correction or continuum theory follows from (13). The scalar is a reference expectation sum, not its interacting ground energy.

The implementations retain positive and negative coupling, zero and empty cases, exact low-order geometry, complete positive tails, wrong factorials, missing quadratic/cross histories, scalar double counting and omitted transport controls. Finite rational matrix fixtures diagnose algebra rather than simulate the full physical Hilbert space. Producer and skeptic constants differ because their sufficient positive majorants differ. The common admitted conclusions are the complete inventory, transport, stated interaction budgets and reference-only bound; scientific priority remains unverified.

4 AG3: one complete correction and its passive comparison**

The second investigation of the homogeneous goal applies a domain-preserving correction to the *complete* actual AG2 mixing family. Its bound is

$$\|R'\|_{3/2} \leq \gamma \|B\|_2, \quad \gamma = \begin{cases} 3060/3623, & 0 \leq M \leq 10^{-3}, \\ 5949/96812, & 0 \leq M \leq 10^{-4}. \end{cases} \quad (14)$$

This compares different support weights. The main coefficient is weaker than the passive bound obtained by leaving B unchanged and measuring it at weight $3/2$. The narrow coefficient improves that universal upper ceiling, but neither interval proves improvement over the actual old lower-weight norm. The exact full conjugation, domains, retained diagonal and scalar terms are substantive parts of the accepted result.

4.1 A bare-reference inverse for every generated support

The actual input from AG2 is

$$H_{\text{in}} = C_{\Lambda}I + H_0 + D_{\text{next}} + B, \quad D_{\text{next}} = D + Z, \quad r = \|B\|_2, \quad d = \|D_{\text{next}}\|_2. \quad (15)$$

Every assigned connected support Y has at least four coarse blocks, with no uniform upper size. All original and transported histories remain indexed. Coincident supports are not canceled or assigned smaller supports to alter r . With the admitted cap K from AG2,

$$r \leq K, \quad d \leq 64M + 2K, \quad |\langle \psi, D_{\text{next}} \psi \rangle| \leq (4M + K/8) \langle \psi, H_0 \psi \rangle. \quad (16)$$

Here $K = 7/1000$ or $7/100000$ on the main or narrow interval, respectively. The actual norm r is distinct from this upper cap and from an extensive operator norm.

Since $B_Y = P_Y K_Y Q_Y + Q_Y K_Y P_Y$, write $v_Y = B_Y \Omega_Y$ and define a new generator T (called X in the AG3 reports):

$$u_Y = (H_{0,Y} |_{Q_Y})^{-1} v_Y, \quad T_Y = |u_Y\rangle \langle \Omega_Y| - |\Omega_Y\rangle \langle u_Y|, \quad T = \sum_Y T_Y. \quad (17)$$

The reduced local reference is at least one. Spectral calculus gives $u_Y \in D(H_{0,Y})$, $H_{0,Y} u_Y = v_Y$, and $\|T_Y\| \leq \|B_Y\|$. The local products have bounded extensions and $[T_Y, H_{0,Y}] = -B_Y$ on the domain. Extend by the exterior identity. Commuting nonnegative local and exterior energies give $D(H_0) = D(H_{0,Y}) \cap D(H_{0,\text{ext}})$, so the full domain is preserved. This is not regularization of an arbitrary exterior vector and is not the operator obtained with a global vacuum projector.

In each finite cuboid,

$$\sum_Y \|B_Y\| \leq |\Lambda| r / 64, \quad \sum_Y \|T_Y\| \leq |\Lambda| r / 64, \quad \|T\|_2 \leq r. \quad (18)$$

Closedness of H_0 and convergence of the bounded commutators give $[T, H_0] = -B$. The relation $H_0 T \psi = T H_0 \psi + B \psi$ supplies the graph-norm bound; both exponentials preserve $D(H_0)$. No bounded infinite-volume global generator or unitary is asserted. This inverse uses H_0 alone, so the changed diagonal is retained as a commutator defect.

4.2 The full infinite remainder with an actual denominator

First integrate the established bounded commutator on $D(H_0)$, then expand only bounded conjugations. The exact result is

$$e^T H_{\text{in}} e^{-T} = C_{\Lambda}I + H_0 + D_{\text{next}} + R', \quad R' = \sum_{n \geq 1} \left(\frac{\text{ad}_T^n D_{\text{next}}}{n!} + \frac{n \text{ad}_T^n B}{(n+1)!} \right). \quad (19)$$

In particular $[T, D_{\text{next}}] + [T, B]/2$ remains at first order; the old scalar commutes but stays in the Hamiltonian. This does not assume a norm-convergent BCH series for unbounded H_0 .

For indexed interactions U, V and $a > b \geq 3/2$, nonempty intersection yields $b^{|Y \cup Z|} \leq b^{|Y|} b^{|Z|}/b$. Retaining both root placements and using $m \exp[-m \log(a/b)] \leq 1/(e \log(a/b))$ proves

$$\|[U, V]\|_b \leq \frac{4}{be \log(a/b)} \|U\|_a \|V\|_a. \quad (20)$$

Every repeated index and complete union is included. The proof imposes no maximum support size and establishes the new output-weight range directly.

At nested order n , allocate ℓ/n per bracket with $\ell = \log(4/3)$. All intermediate weights are at least $3/2$. The elementary bound $n! \geq (n/e)^n$ gives

$$\frac{\|\text{ad}_T^n V\|_{3/2}}{n!} \leq (\bar{c}r)^n \|V\|_2, \quad \bar{c} = \frac{288}{31}. \quad (21)$$

Indeed $\ell \geq 31/108$ follows by integrating $1/(1+t) = 1 - t + t^2 - t^3 + t^4/(1+t)$ on $[0, 1/3]$; its remainder is positive. Therefore $(8/3)/\ell \leq \bar{c}$. With $\theta = \bar{c}r < 1$, the complete positive tail yields

$$\|R'\|_{3/2} \leq r \frac{\bar{c}(d+r)}{1-\bar{c}r}, \quad \text{tail after order } N \leq (d+r) \frac{\theta^{N+1}}{1-\theta}. \quad (22)$$

Only the increasing coefficient uses $r \leq K$ and $d \leq 64M + 2K$; the actual r remains outside. At the main and narrow endpoints $\theta \leq 252/3875$ and $63/96875$, respectively. These positive monotonicities establish continuous-interval coverage and give (14). If $r = 0$, each indexed B_Y vanishes, $T = 0$ and $R' = 0$, without division; both coupling signs are covered by $M = 7|\tau|$.

4.3 What passive reweighting already provides

Minimum support size four gives, for the *same* indexed input family,

$$\|B\|_{3/2} \leq (3/4)^4 \|B\|_2 = \frac{81}{256} r. \quad (23)$$

Exact comparison places 3060/3623 above this ceiling and 5949/96812 below it. The main active certificate consequently does not certify a benefit over doing no correction at the lower weight. The narrow active certificate is smaller than the universal passive upper ceiling, but two upper bounds do not compare the actual norms. A diagnostic family concentrated on one support of size m has passive ratio $(3/4)^m$, arbitrarily small as m grows. This is a counterexample to the proposed general inference, not an evaluation of the actual AG2 trajectory.

4.4 Retained scalars, diagonals and the reference-only gap

Split every generated self-adjoint indexed residual J_Y on its full support:

$$\begin{aligned} c'_Y &= \langle \Omega_Y, J_Y \Omega_Y \rangle, & B'_Y &= P_Y J_Y Q_Y + Q_Y J_Y P_Y, \\ Z'_Y &= Q_Y (J_Y - c'_Y I) Q_Y, & J_Y &= c'_Y I + B'_Y + Z'_Y. \end{aligned} \quad (24)$$

The respective norm costs are one, one and two. Put $W = \gamma K$ and $C' = \sum_Y c'_Y$. Retaining the old scalar and minimum assigned support gives

$$\begin{aligned} \|B'\|_{3/2} &\leq \gamma r \leq W, & \|Z'\|_{3/2} &\leq 2W, \\ \frac{|C_\Lambda + C'|}{|\Lambda|} &\leq \frac{K}{64} + \frac{4W}{81}, & |\langle \psi, Z' \psi \rangle| &\leq \frac{32W}{81} \langle \psi, H_0 \psi \rangle. \end{aligned} \quad (25)$$

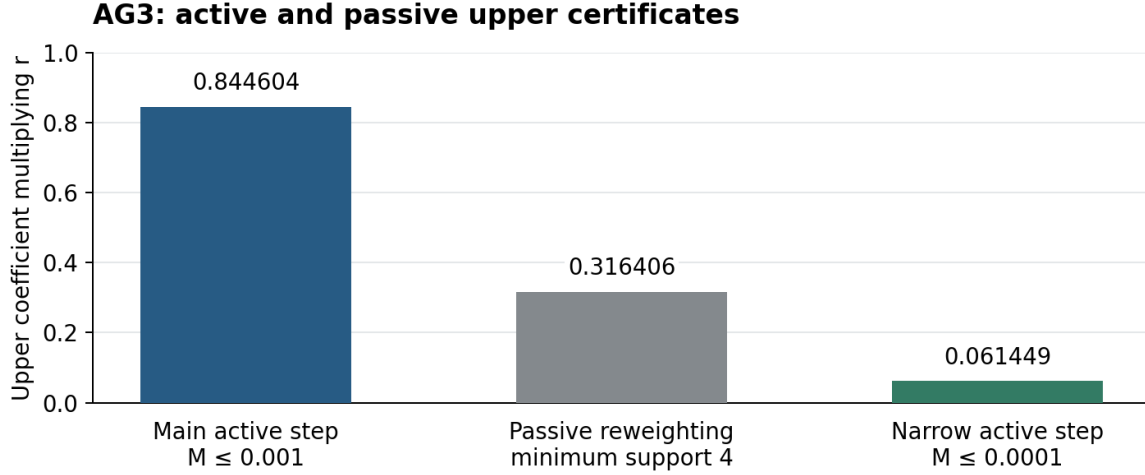


Figure 1: Sufficient coefficients multiplying the same actual $r = \|B\|_2$. Active bars bound the complete corrected $\|R'\|_{3/2}$; the passive bar bounds the unchanged $\|B\|_{3/2}$. All values are exact rational certificates rendered as decimals. They do not evaluate either actual lower-weight norm, establish same-weight contraction, or measure theorem completion.

The extracted centered reference $G' = H_0 + D_{\text{next}} + Z'$ has domain $D(H_0)$, annihilates the product vacuum, and satisfies

$$G' \geq \left(1 - 4M - \frac{K}{8} - \frac{32W}{81}\right) H_0. \quad (26)$$

The exact endpoint coefficient is $258975047/260856000 > 124/125$ on the main interval and $174190074443/174261600000 > 1999/2000$ on the narrow interval. These are normalized *reference-only* gap floors. Physical reference energies multiply the unchanged $\delta = \alpha/8$. The full corrected operator remains $(C_\Lambda + C')I + G' + B'$. Its mixing is not discarded, its scalar is not identified with the true interacting ground energy, and its gap is unproved.

The reverse derivation additionally applies passive reweighting to the carried diagonal, obtaining $\|D_{\text{next}} + Z'\|_{3/2} \leq (81/256)d + 2W$. The forward and skeptical bound $d + 2W$ is also valid. This difference in sufficient estimates is preserved rather than silently merged.

The bare-inverse strategy originated in the Newton post-AG2 review; the advisor's overlap refinement and proposed constant were shared before the contract. The directions independently derived and implemented the frozen target, not independently invented its premise. The skeptic's independent production preceded exchange; its additional passive interpretation and the reverse supplemental note were post-freeze reviews. Exact finite-matrix, exterior-domain, complete-support and denominator controls test the stated algebra and inference. They are not physical $SU(2)$ cutoff simulations or observations.

Both the previous $9/4 \rightarrow 2$ and current $2 \rightarrow 3/2$ margins are spent. Growing-support controls reject a free return to weight two, and a generic fixed-weight commutator control rejects a universal bilinear constant for that broad class. No all-stage conclusion or repair of the earlier O2 majorant follows. A future iteration requires its own remaining weight schedule, retained-diagonal estimates and convergence theorem. This loop proves no full homogeneous gap, physical scale matching or four-dimensional continuum construction.

5 AI3: exact retained cancellation and full physical readout limits**

AI3 calculates exact odd moments and proves an all-order retained cancellation lemma. The complete physical ratio intervals nevertheless overlap in all ten frozen cells. The direct scalar distinction at $u = 10^{-18}$, $k = 5$ survives for both actual Wilson multipliers. These conclusions concern sufficient certificates in the canonical summable full-link SU(2) family, under its inherited transfer theorems. They do not prove equality of the candidates, a general identification impossibility or a homogeneous-model result.

5.1 Actual multipliers, complete factors and inherited transfer

Use the free unit cube based at $(3, 1, 0)$ and the closed six-link path

$$X : (3, 1, 0), (3, 2, 0), (3, 2, 1), (4, 2, 1), \\ (4, 2, 0), (4, 1, 0), (3, 1, 0).$$

The two yz faces f_4, f_5 are anchored at $(3, 1, 0)$ and $(4, 1, 0)$. Their symmetric differences with X give simple six-cycles $Y_r = X \triangle f_r$ and the actual multiplication observables

$$B_r = \frac{\chi_X + \chi_{Y_r}}{\sqrt{2}}, \quad \psi_r = B_r \Omega = \frac{e_X + e_{Y_r}}{\sqrt{2}}, \quad r = 4, 5. \quad (27)$$

An unmatched free-link center action gives zero means and orthogonality, so $\|\psi_r\| = 1$. Conditional Haar integration over exclusive paths gives $\langle B_r^4 \rangle = 5/2$. The exact multiplication norm is $J = 2\sqrt{2}$, with its supremum approached on positive-measure neighborhoods of identity holonomies. Therefore $\|(B_r^2 - I)\Omega\|^2 = 3/2$: a normalized source does not authorize a norm-one multiplier or rank-source replacement.

Each probe occupies eight links and their union occupies ten free factors. Independent cube enumeration reconstructs all sixteen six-cycles and the reached face-flip component at reference energy $E = 9\alpha/2$. This component reduces the pinched average; it is not the full regional electric shell or an invariant subspace of the unaveraged dynamics. The twenty exterior touching faces are explicitly retained. Each shares one cube edge, has a free opposite edge and two distinct external complete-factor owners. The possible 3/4 saving on the shared edge is canceled by the opposite free-edge cost; the side owners cost at least 1/8 each. With disjoint and internal nonresonant cases, this verifies applicability of the inherited participating-frequency bound 1/8 to both probes. The infinite-dimensional domains, strip gaps, pinching and strong averaging theorem remain inherited premises from Y2/AA2. AD2 supplies the correctly ordered scalar identity

$$e^{iEt/\hbar} \langle \Omega, B_r \beta_q^t(B_r) \Omega \rangle = \langle \psi_r, Z_q(s) B_r Z_q(s)^* \Omega \rangle. \quad (28)$$

Both source and adjoint-vacuum evolution errors are charged. No finite matrix enumeration substitutes for these full-link transfer theorems.

Starting from the common ten-factor seed, each collar adjoins every owner of every omitted face touching complete current factors. A ten-link strip is always included whole. A face is retained only when all its owners are included. The exact reconstructed counts are

Depth k	Complete factors	Complete links	Retained faces N_k
0	10	10	2
1	41	113	25
2	149	329	143
3	333	675	359
4	610	1177	695
5	993	1857	1171
6	1497	2739	1808

The depth-six extension was declared before production; it has 813 crossing faces. Complete links and owners, rather than counts alone, are exported. The old eight-link seed fails the second probe's support requirement.

At order n , connected words visit at most $10 + 3n$ factors, each with at most forty incident faces. The interaction coefficient, commutator factor two and time simplex give $(10/3)_n x^n / n!$, where $x = 10\alpha\tau|t|/\hbar$. All full and regional words through order k agree. A complete positive tail, including repetitions and outward-return paths, is bounded at $x \leq 15z < 1$ by

$$D_k^{(10)}(15z) = \frac{(10/3)_{k+1}}{(k+1)!} \frac{(15z)^{k+1}}{(1-15z)^{k+5}}. \quad (29)$$

The integer denominator exponent conservatively rounds the positive Taylor remainder exponent $k + 13/3$ upward. This is a spatial estimate for complete infinite-dimensional factors, not a representation cutoff.

5.2 Exact moments and a reflection proof at every retained order

The finite reached matrix $Q(q)$ places weights q^4 or q^5 on physical face flips. Its symmetric row sums give $\|Q\| \leq 6$ and $\|Q'\| \leq 30$ on $[0, 1]$. Set

$$p(q) = 2 + 5q + 5q^2 + 6q^3 + 3q^4, \quad v = \frac{z(1+q)^2(1+q^2)}{32p(q)},$$

$$h_r = \langle \psi_r, e^{ivQ(q)} \psi_r \rangle, \quad m_{n,r} = \langle \psi_r, Q(q)^n \psi_r \rangle, \quad \Delta_n = m_{n,5} - qm_{n,4}. \quad (30)$$

Exact polynomial walks, separately checked against rational matrix propagation, give

$$\begin{aligned} \Delta_1 &= 0, & \Delta_3 &= 2q^{13}(1-q^2), \\ \Delta_5 &= q^{21}(1-q^2)(12+8q+12q^2), \\ \Delta_7 &= q^{29}(1-q^2)(54+72q+164q^2+72q^3+54q^4). \end{aligned} \quad (31)$$

Let P_r denote the degree-eight exponential center. The alternating imaginary signs are retained:

$$C_8 := \text{Im } P_5 - q \text{Im } P_4 = -\frac{v^3 \Delta_3}{3!} + \frac{v^5 \Delta_5}{5!} - \frac{v^7 \Delta_7}{7!}. \quad (32)$$

In particular the cubic contribution is negative for $0 < q < 1$.

Reflection $(x, y, z) \mapsto (7-x, y, z)$ fixes the unoriented cycle X , swaps Y_4, Y_5 , and induces an involution on all sixteen cycles. Every unweighted face-flip adjacency is preserved, so its permutation P obeys $PQ(1) = Q(1)P$. Fundamental $\text{SU}(2)$ characters are unchanged by reversal. Thus $P\psi_4 = \psi_5$ and $\Delta_n(1) = 0$ for *every* n . This is an all-order retained proof, not an extrapolation from finitely many moments. The reflection does not preserve all anchored weights at $q < 1$ and supplies no full canonical $q = 1$ Hamiltonian.

At fixed normalized sources, the product rule gives

$$|m'_{n,r}(q)| \leq 5n6^n, \quad |\Delta'_n(q)| \leq (10n+1)6^n, \quad |\Delta_n(q)| \leq (1-q)(10n+1)6^n. \quad (33)$$

The last inequality integrates the derivative from q to one after the reflection proof. No physical error disk or parameter-dependent physical correlation is differentiated; $v(q)$ is substituted afterward. Since $(10n+1)/n!$ decreases for $n \geq 9$, for $x = 6|v| < 1$ the whole retained tail obeys

$$|\text{Im } h_5 - q \text{Im } h_4 - C_8| \leq T_\Delta := \frac{(1-q)91x^9}{9!(1-x)}, \quad |h_r - P_r| \leq A := \frac{x^9}{9!}. \quad (34)$$

Even powers are included conservatively in the first sum. Under the first hypothesis at $u = 10^{-18}$, A is about 1.334×10^{-70} while T_Δ is about 1.214×10^{-86} . The new retained arithmetic estimate does not reduce physical uncertainty.

5.3 Physical errors, shared denominators and valid enclosures

The unchanged canonical quantities are

$$\begin{aligned} b(q) &= \frac{p(q)}{24(1-q)^3(1+q)^2(1+q^2)}, & \tau_q &= \frac{\eta}{8b(q)}, \\ \hat{d}_q &= \frac{\tau_q \sqrt{b(q^2)/96}}{(1-\eta)/8}, & M_k &= \frac{N_k}{24}, \\ R_{\text{phys}} &= 48\hat{d}_q + 8D_k^{(10)}(15z) + 64\tau_q M_k(1 + 3zM_k). \end{aligned} \quad (35)$$

The first term retains stationary connected-state replacement; the spatial factor is $J^2 = 8$; the averaging term bounds $(1+J)16\tau_q M_k(1 + 3zM_k) < 64\tau_q M_k(1 + 3zM_k)$ and retains both evolution columns. Exact upward square-root enclosures differ between implementations but preserve the verdicts. Unsupplied preparation, measurement, phase and clock errors must be added separately. Write $a_r = \text{Im } P_r$ and $y_r = a_r + \delta_r + e_r$, with $|\delta_r| \leq A$ and $|e_r| \leq R_{\text{phys}}$. Every frozen denominator interval

$$I_4 = [a_4 - R_{\text{phys}} - A, a_4 + R_{\text{phys}} + A] \quad (36)$$

has positive lower endpoint. All-corner division of the corresponding I_5 by I_4 gives a valid direct ratio enclosure, including signed numerator cases. A second valid construction divides $C_8 \pm [(1+q)R_{\text{phys}} + T_\Delta]$ by I_4 and adds q . Its rectangular relaxation forgets the shared e_4 in numerator and denominator. Equal physical radii do not imply equal signed errors: $e_5 = R_{\text{phys}}$ and $e_4 = -R_{\text{phys}}$ attain the full $(1+q)R_{\text{phys}}$ combination allowance.

The reverse derivation additionally keeps that dependence in the enlarged box

$$q + \frac{C_8 + \delta_C + e_5 - qe_4}{a_4 + \delta_4 + e_4}, \quad |\delta_C| \leq T_\Delta, \quad |\delta_4| \leq A, \quad |e_4|, |e_5| \leq R_{\text{phys}}. \quad (37)$$

With positive denominator, each coordinate is linear-fractional and monotone or constant when the others are fixed, so its sixteen corners give exact extrema of this enlarged box. This is not a claim that all corners are physically attainable. Intersections of valid enclosures remain valid. Smaller width does not establish endpoint containment; no blanket containment between the direct and cancellation intervals is asserted, and no global optimality over all retained constraints is claimed. The final review finds smaller direct interval widths in all twenty hypotheses, but none of those direct intervals is wholly contained in its cancellation-centered counterpart. The forward post-freeze clarification explicitly preserves this distinction.

The measured statistic y_5/y_4 contains no unknown q . A residual $y_5 - q_c y_4$ is explicitly indexed by a proposed candidate q_c ; the unknown true parameter cannot be supplied secretly as an instrument setting. The old cubic comparator, with center vq^r and additional floor $36v^3$, is recalculated on the same grid with complete errors and denominator checks.

5.4 Ten insufficient ratio cells and a surviving scalar distinction

With $z = 10^{-6}$, the two candidates are

$$H_1 : (q, \eta) = (1 - u, 1/2), \quad H_2 : (q, \eta) = (1 - 2u, 1/16). \quad (38)$$

Both have $\eta(1-q)^3 = u^3/2$ and the same original physical time

$$t_u = 2 \cdot 10^{-6} u^{-3} \frac{\hbar}{\alpha} = \begin{cases} 2 \cdot 10^{30} \hbar/\alpha, & u = 10^{-12}, \\ 2 \cdot 10^{48} \hbar/\alpha, & u = 10^{-18}, \\ 2 \cdot 10^{66} \hbar/\alpha, & u = 10^{-24}. \end{cases} \quad (39)$$

Positive α , $\hbar$, α/E_* and spacing remain fixed. No independent clock adjustment for the candidates is permitted.

For two ratio intervals $[l_i, u_i]$, the signed separation margin is $\max(l_1 - u_2, l_2 - u_1)$. Positive means certified disjointness. Every direct full-error margin below is negative; cancellation-centered, shared-corner and intersected enclosures also remain insufficient in all ten cells. Exact fractions determine signs; decimals only display scale.

u	k	Direct ratio margin (approx.)	Full scalar tests B_4, B_5
10^{-12}	3	$-3.811493484 \cdot 10^{-9}$	insufficient
10^{-12}	4	$-1.044822364 \cdot 10^{-9}$	insufficient
10^{-12}	5	$-1.044761496 \cdot 10^{-9}$	insufficient
10^{-18}	3	$-2.766731990 \cdot 10^{-9}$	insufficient
10^{-18}	4	$-6.086906259 \cdot 10^{-14}$	insufficient
10^{-18}	5	$-1.313885034 \cdot 10^{-18}$	both disjoint
10^{-24}	3	$-2.766731990 \cdot 10^{-9}$	insufficient
10^{-24}	4	$-6.086901682 \cdot 10^{-14}$	insufficient
10^{-24}	5	$-1.268122540 \cdot 10^{-18}$	insufficient
10^{-24}	6	$-2.436389698 \cdot 10^{-23}$	insufficient

Replacing the separate cubic linearizations by degree-eight centers removes the material old Taylor floor, approximately $2.0408163 \times 10^{-14}$ in the ratio comparison, and sharpens the last cells. Ordinary degree-eight arithmetic is already negligible on this grid. The further all-order $(1 - q)$ factor improves that negligible arithmetic term; it does not cause a successful ratio discrimination or reduce the physical upper budgets. Overlap of sufficient intervals is not a lower bound on actual error, physical equality or an impossibility theorem.

Component-only recalculations diagnose these bounds and are never substituted for the full radius. At $u = 10^{-18}$, $k = 5$, the state-only ratio margin is approximately $-4.576149531 \times 10^{-20}$ and the spatial-only margin is $-2.681235390 \times 10^{-19}$. Either inherited component alone leaves this interval test insufficient. The two complete scalar radii are about $9.011406565 \times 10^{-27}$ and $4.761718639 \times 10^{-27}$, combining unequal state costs with the common spatial cost $3.774177199 \times 10^{-27}$.

At $u = 10^{-24}$, $k = 6$, the spatial-only ratio margin is approximately $-2.436285122 \times 10^{-23}$, while the state-only margin is positive, about $9.989542385 \times 10^{-25}$. Each spatial scalar budget is $7.548467626 \times 10^{-32}$; the state budgets are about $5.237229366 \times 10^{-36}$ and $9.875414398 \times 10^{-37}$. This isolates the spatial certificate as the relevant remaining bound at the predeclared depth; it does not measure the actual dominant error.

The surviving direct scalar margins at $u = 10^{-18}$, $k = 5$ are about $3.214524214 \times 10^{-26}$ for B_4 and $4.405000405 \times 10^{-26}$ for B_5 . Equal additional absolute error on each compared scalar may be strictly below half the respective exact margin. These are pairwise mathematical allowances at time $2 \times 10^{48} \hbar/\alpha$, not instrument sensitivity. No ratio cell has a positive margin, so no new ratio tolerance is inferred.

The common *known* reference carrier in (28) can be removed consistently. A common unknown phase generally changes an imaginary-part ratio even though it preserves complex disk distances. Independently fitted phases can also change disk distances. Thus unspecified phase, timestamp, preparation and measurement errors are not canceled, and no timing-sensitivity theorem is claimed.

The Tesla-method proposal supplied the candidate-indexed readout strategy and the advisor supplied the prospective reflection lemma. The directions and skeptic independently derived consequences of this shared context. Complete geometry, exact moment signs, finite-moment counterexamples, actual multiplier norms, omitted-vacuum-column controls, signed interval corners and phase rotations test the argument. These are mathematical controls rather than observations. No continuous inverse, realized apparatus, homogeneous matching map, nontrivial four-dimensional

continuum construction or Yang–Mills mass-gap theorem follows; scientific priority remains unverified.

6 AI4: a uniform family of complete physical ratio tests^{**}

AI3 identified a remaining spatial certificate at small u . AI4 adaptively shortens the prescribed duration to $z = 10^{-7}$ and increases the proof collar with a dyadic schedule. Under the inherited canonical full-link transfer theorems, the two complete predicted ratio intervals have gap at least $39u/40$ for *every real* $0 < u \leq 10^{-24}$. Both independent producers prove this common conclusion using different explicit upper bounds. Every state, spatial and averaging error remains; this is a conditional family of candidate-pair tests, not a continuous inverse or an apparatus.

6.1 The family and its unchanged physical clock

Fix $u_0 = 10^{-24}$, $z = 10^{-7}$ and let $j \geq 0$ be the unique integer for which

$$\frac{U_j}{2} < u \leq U_j, \quad U_j = u_0 \cdot 2^{-j}, \quad k = 6 + j. \quad (40)$$

An excluded lower endpoint belongs to the next band. The candidates remain (38); hence $\eta(1 - q)^3 = u^3/2$ in both models and their common physical time is

$$t = 2zu^{-3} \frac{\hbar}{\alpha}. \quad (41)$$

At the base upper endpoint it is already $2 \times 10^{65} \hbar/\alpha$ and grows without bound. Positive α , $\hbar$, α/E_* and spacing remain fixed. Changing z changes the observation duration within the admitted range, not the Hamiltonian or a separately fitted candidate clock. The values q, η are canonical-model parameters, not lattice spacing.

The actual probes, multiplication norm $2\sqrt{2}$, common ten-factor seed, reached sixteen-cycle component and exterior-loading hypotheses are those of AI3. The complete physical radius remains (35), now at the shorter duration and the scheduled k . Both connected-state means, both evolution columns, complete factors, and every omitted spatial word remain. The reached averaged component is not the whole electric shell or an instantaneous invariant space. The following proof uses source-bound AA2/AD2/Y1/Y2/AI3 transfer premises; it does not replace them with matrices or revalidate the entire historical proof chain.

6.2 A collar envelope proved at every depth

A unit face touching a current link adds at most one to each endpoint-box coordinate. Completing a strip owner adds at most three in x , one in y and zero in z ; its actual six x links and four y links are all included. A free factor adds no completion. Induction on the complete owner rule therefore gives

$$\text{vertices}(E(F_k)) \subset [0, 4(k+1)] \times [0, 2(k+1)] \times [0, k+1] \quad \text{for every } k \geq 0. \quad (42)$$

Negative-coordinate faces are absent. Every retained face lies wholly inside this box. Counting unit faces in its three orientations yields the forward envelope, with $L = k + 1$,

$$N_k \leq \overline{N}_k = 24L^3 + 14L^2, \quad M_k \leq \overline{M}_k = L^2(L + 7/12). \quad (43)$$

The reverse proof instead counts three orientations at every possible anchor, obtaining the valid larger bound

$$N_k \leq 3(4k+5)(2k+3)(k+2), \quad M_{6+j} \leq 435(j+1)^3. \quad (44)$$

At depth six, the exact admitted face count is 1808, while the two envelopes are 8918 and 10440. Every admitted depth-zero-through-six complete link is checked against the new geometry. No actual collar beyond depth six is enumerated; a large-depth polynomial envelope is never presented as an exact face count.

Hereafter the displayed detailed proof follows the forward constants. Put

$$M_j = (j+7)^2(j+7+7/12), \quad M_0 = 4459/12, \quad x = 15z. \quad (45)$$

The full spatial tail in (29) satisfies

$$\frac{D_{k+1}}{D_k} = \frac{x}{1-x} \frac{k+13/3}{k+2}, \quad \frac{D_{6+j}}{u} \leq \frac{2^{j+1}D_{6+j}}{u_0} \leq \frac{2D_6}{u_0}, \quad (46)$$

because the successive ratio of the middle majorants is at most $2x(31/24)/(1-x) < 1$. This proves every integer j , rather than extrapolating from sampled bands.

Each of the three positive affine factors in M_j grows by at most $8/7$ per step. Thus $M_{j+1} \leq gM_j$, with $g = (8/7)^3$, and

$$\frac{g^2}{4} < 1, \quad \frac{g^2}{8} < 1. \quad (47)$$

Consequently both $U_j^2 M_j(1+3zM_j)$ and $U_j^3 M_j(1+3zM_j)$ decrease. The full quadratic M_j^2 term is retained. Since both candidates obey $\tau_q \leq 3u^3/4$, the averaging contribution is bounded by $48u^3 M_j(1+3zM_j)$, with its normalized and absolute maxima controlled at the base band. The reverse proof independently bounds the cubic and sixth-power sequences $(j+1)^3/4^j \leq 2$ and $(j+1)^6/4^j \leq 64$ by finite algebraic prefixes followed by strict ratio induction. Those prefixes are not additional physical fixtures.

6.3 Continuous bounds and the location of arithmetic error

For the entire family $q \geq q_* = 99/100$. With $e = 1 - q$, positive polynomial inequalities for the unchanged $p(q), b(q)$ give

$$\frac{z}{100} \leq v \leq V := \frac{z}{64}, \quad \tau_q \leq \frac{3}{2}\eta e^3, \quad \sqrt{b(q^2)/96} \leq \frac{1}{80e^{3/2}}. \quad (48)$$

For example $p(q) - 2(1+q)^2(1+q^2) = q + q^2 + 2q^3 + q^4 \geq 0$ gives the upper v bound; rational endpoint inequalities at q_* give the others. No grid interpolation supplies these continuous claims.

The complete forward state bounds for H_1, H_2 are $(36/5)u^{3/2}$ and $(36/25)u^{3/2}$. An absolute mesh $\delta = 2^{-320}$ applied to the *inner* square root adds at most

$$\frac{384\tau_q\delta}{1-\eta}, \quad \text{hence at most } 576\delta u^3 \text{ or } (1536/5)\delta u^3. \quad (49)$$

The prefactor is essential. A fixed uncertainty added to the final scalar is a different error and does not gain this cubic decay. The uniform proof for arbitrary real u needs no numerical square root; the propagated allowance remains conservative, and the exact rational fixtures check its location. The reverse proof uses different analytic state bounds, $9u^{3/2}$ and $(9/5)u^{3/2}$, with no numerical root needed in its theorem.

AI3's factored moments imply $|\Delta_3| \leq 4e$, $|\Delta_5| \leq 64e$, $|\Delta_7| \leq 832e$. Define exact rational constants

$$B_* := \frac{2}{3}V^3 + \frac{8}{15}V^5 + \frac{52}{315}V^7, \quad T_* := \frac{91(6V)^9}{9!(1-6V)}, \quad A_* := \frac{(6V)^9}{9!}. \quad (50)$$

The complete retained combination satisfies $|C_8| \leq eB_*$ and its remaining tail is at most eT_* . Only fixed-source retained moment polynomials enter this statement; no physical error is canceled or

differentiated. The ordinary individual allowance A_* can remain constant in a positive denominator. Replacing the vanishing numerator allowance by a constant individual Taylor error would not establish this family for arbitrarily small u .

Nonnegative retained moments and alternating center signs give

$$a_4 \geq d_{\text{poly}} := \frac{zq_*^4}{100} - 36V^3 - \frac{(6V)^7}{7!}. \quad (51)$$

The positive fifth-order contribution is dropped only for this lower bound. Let

$$R_* := \frac{36}{5}u_0^{3/2} + 576\delta u_0^3 + 8D_6 + 48u_0^3M_0(1 + 3zM_0). \quad (52)$$

The continuous bounds and all-band recurrences above imply $R_h \leq R_*$ for both candidates. Exact rational evaluation proves

$$d_{\text{poly}} - A_* - R_* > d_* := z/120 > 0, \quad |y_5| < n_* := z/32. \quad (53)$$

This bounds actual full scalars and the exported scalar-center denominator intervals, not just the retained matrix denominator.

6.4 Explicit uniform gap and the independent constants

Physical errors remain adversarially bounded:

$$|y_5 - qy_4| \leq e(B_* + T_*) + (1 + q)R_h \leq e(B_* + T_*) + 2R_h. \quad (54)$$

For $h = 1, 2$, put $(a_1, a_2) = (36/5, 36/25)$ and $(b_1, b_2) = (576, 1536/5)$, and define six exact contributions

$$\begin{aligned} c_h^{\text{state}} &= \frac{2a_h\sqrt{u_0}}{d_*}, & c_h^{\text{round}} &= \frac{2b_h\delta u_0^2}{d_*}, \\ c_h^{\text{spatial}} &= \frac{32D_6}{d_*u_0}, & c_h^{\text{average}} &= \frac{96u_0^2M_0(1 + 3zM_0)}{d_*}, \\ c_h^{\text{bias}} &= \frac{hB_*}{d_*}, & c_h^{\text{tail}} &= \frac{hT_*}{d_*}. \end{aligned} \quad (55)$$

The band lower endpoint controls the spatial term; the upper endpoint controls the remaining physical terms. With c_h their sum, every real u in every band obeys

$$\frac{y_5}{y_4} \in [q_h - c_h u, q_h + c_h u], \quad \text{gap} \geq \beta u, \quad \beta := 1 - c_1 - c_2 \approx 0.9791915454711759 > 39/40. \quad (56)$$

The definition makes β an explicit rational constant after the frozen substitutions; its full fraction is retained in the output. It is a sufficient coefficient, not an optimal physical margin or a measured signal-to-noise ratio.

The reverse producer obtains a different valid coefficient $\Gamma \approx 0.975932912473311 > 39/40$ using its envelope and denominator $d > z/125$. Its compact formula is

$$\Gamma = 1 - \frac{3(B_* + T_*) + 2(P_1 + P_2)}{d}, \quad (57)$$

where P_h are its complete physical-radius coefficients, $R_h \leq uP_h$. These include the $9, 9/5$ state constants, the full spatial tail, and the proved growing-collar averaging bound. Its actual d is the explicit rational lower bound from its scalar-center estimate, not the relaxed number $z/125$. The independent skeptic obtains a third sufficient coefficient near 0.9753467151692543 . These

are independently derived certificates with different conservative choices; the common public conclusion is the weaker $39u/40$ floor.

In the forward bound the summed deficit is dominated by state, 0.020736, and spatial, approximately 7.24545×10^{-5} . Averaging, correctly propagated rounding, retained bias and the full retained tail are also charged. Shortening duration reduces the spatial certificate strongly while increasing the relative state cost; the new schedule succeeds only because the combined bound remains below one.

6.5 Vanishing scalar tolerance, exact fixtures and limits

For an additional absolute error at most ν in each scalar, (53) gives a ratio change at most

$$L\nu, \quad L = \frac{2}{d_*}(1 + n_*/d_*) = \frac{19}{2d_*}, \quad \nu \leq d_*/2. \quad (58)$$

The explicit forward choice

$$\nu(u) = \frac{d_*u}{100} = \frac{zu}{12000} \quad (59)$$

charges both candidates and leaves gap at least $(\beta - 19/100)u > u/2$. At u_0 the allowance is about 8.33333×10^{-36} . It decreases to zero with u : no fixed positive apparatus error is covered uniformly. The four accepted allowances have distinct denominators and remaining margins:

Derivation	Additional error per scalar	Remaining gap
Forward	$d_*u/100$	$(\beta - 19/100)u > u/2$
Reverse, half margin	$\Gamma du/24$	$\Gamma u/2$
Reverse, target margin	$(\Gamma - 1/2)du/12$	$u/2$
Skeptic, $D = z/128$	$39Du/1280$	$39u/80$

Each denominator reserve is proved in its own derivation. No mixed allowance/margin theorem is asserted, and the half-margin choices need not preserve $u/2$. Supplied preparation, measurement or calibration uncertainty must fit the applicable total scalar allowance.

Exactly ten physical parameter fixtures are evaluated: $j \in \{0, 1, 8, 64, 512\}$, each at $u = U_j$ and $u = 3U_j/4$. They retain exact $q < 1$, rational centers, complete errors and the proved face envelopes. At the two existing $j = 0$ points only, the admitted exact $N_6 = 1808$ is compared with the envelope; this is not an extra cell. No new collar beyond six is computed. Large-band decimal displays that round a normalized gap to one do not erase its exact nonzero deficit.

Fixture success checks implementation. The theorem's continuous and all-integer quantifiers come from the geometry induction, continuous parameter inequalities and recurrences above. Separate algebraic controls reject finite-index extrapolation, wrong endpoint assignment, incomplete strip support, a fixed collar count, dropping M^2 , deleting physical error terms, a wrong candidate clock, and confusing an inner arithmetic allowance with a final scalar floor. They do not supply extra physical observations or an automatic enlarged-graph Gram or omitted-channel theorem.

The shorter-duration strategy was proposed in the Feynman-method review after AI3; the advisor proposed its dyadic extension. These shared premises are attributed, while the producers and skeptic independently establish their consequences. The measured ratio contains no unknown q ; candidate-dependent residuals and the time schedule are predictions for predeclared pairs. This is not a universal instrument setting secretly supplied an unknown u , or an inverse from fixed unrestricted data.

The common reference carrier and physical scale must be independently fixed. An unknown common phase need not preserve an imaginary-only ratio, and unsupplied timestamp or phase error requires a separate sensitivity estimate. There is no practical clock, apparatus, fixed-error robustness, homogeneous-model matching, global $q = 1$ Hamiltonian, continuum construction or Yang–Mills mass-gap theorem. Scientific priority remains unverified.

7 AH1: complete physical enrichment on a larger open graph**

AH1 reconstructs the open $3 \times 2 \times 1$ -cell graph, its strict physical electric projection and every individually resolved channel in the specified enrichment. The graph has 24 vertices, 46 links and 29 faces, including seven internal faces. The initial projection has rank 30; the enriched physical space has rank 561, a positive exact Haar Gram and 2,124 nonzero retained magnetic entries. For the contracted preparation class, the exact full and retained heat evolutions, each centered at its own actual ground energy, have all-time true-output-relative error below $11/5000 = 0.0022$. This is an analytic certificate, not a numerical heat trajectory or a graph-size-uniform result.

7.1 Every original link, boundary gauge action and full domain

The vertex set is $\{0, 1, 2, 3\} \times \{0, 1, 2\} \times \{0, 1\}$. Links are oriented in positive coordinate directions; a face with ordered axes $a < b$ follows the word $+a, +b, -a, -b$. The complete list has twelve xy , nine xz and eight yz faces. Its cycle rank $46 - 24 + 1 = 23$ is a graph invariant, not the dimension of a linear Wilson-function basis. All boundary vertices carry the gauge action $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$; a closed face word transforms by conjugation at its base. No tree links or original electric derivatives are deleted.

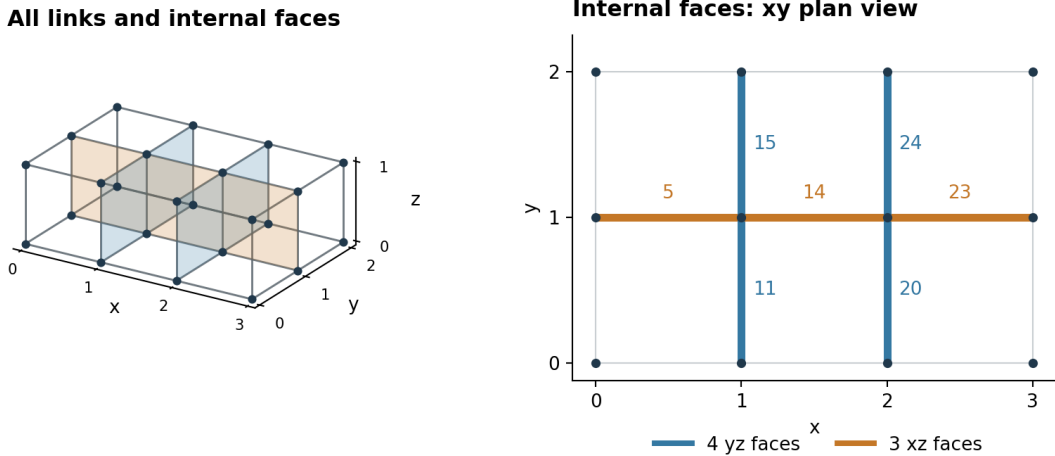


Figure 2: The actual AH1 coordinate graph and its seven internal faces. The left panel retains every one of the 46 links and 24 vertices. The right panel projects the seven vertical internal faces onto the xy plane; numbers are frozen face IDs. Colors distinguish the two plane orientations. This is graph geometry, not a proposed physical apparatus or a heat simulation.

On normalized product Haar measure, the actual model is

$$\begin{aligned} \mathcal{H}_{\text{phys}} &= L^2(\text{SU}(2)^{46})^{\text{SU}(2)^{24}}, & K &= \sum_e C_e, \\ W_p &= \frac{1}{2} \text{Tr } U_{\partial p}, & S &= \sum_{p=0}^{28} W_p, & L &= K + \lambda(29 - S). \end{aligned} \quad (60)$$

Product Peter–Weyl labels give electric energies $\sum_e j_e(j_e + 1)$ with finite multiplicities below every bound. Gauge averaging preserves each complete label block and commutes with the Casimirs. The restricted K is self-adjoint with compact resolvent, operator domain the invariant product H^2 space, form domain invariant H^1 , and smooth invariant Peter–Weyl core. Since $0 \leq 29 - S \leq 58$, bounded perturbation gives $D(L) = D(K)$ and the same form domain and compact resolvent. For the declared $0 \leq \lambda \leq 1/100$, $L \geq K$. These are full representation-space statements, not consequences of a finite matrix cutoff.

Physical energy is $H = \alpha L$, and heat time is $\sigma = \alpha t / \hbar$, with $a, E_*, \alpha / E_*, \hbar$ fixed and positive. This is a finite uniform-coupling graph model, distinct from the preceding homogeneous and canonical families. The theorem concerns Euclidean heat, not real-time evolution.

7.2 The strict electric cutoff and actual Haar quotient

Every nontrivial invariant spin support has no degree-one active vertex, and every active edge costs at least $3/4$. Below $9/2$ at most five edges are active. A nonempty such support contains a cycle; the graph is simple, bipartite and of girth four. Two components would require at least eight edges. A five-vertex, five-edge minimum-degree-two component would be an odd five-cycle, while a bipartite support on at most four vertices has at most four edges. Hence the only possible nonempty support is a four-cycle. Complete graph enumeration identifies these exactly with the 29 elementary faces.

At bivalent vertices the invariant tensor forces a common spin around the cycle. The strict inequality $4j(j+1) < 9/2$ selects $j = 1/2$, and its normalized function is $\phi_p = 2W_p$. Therefore

$$P_0 = \mathbf{1}_{[0,9/2)}(K), \quad \text{ran } P_0 = \text{span}\{\Omega, \phi_0, \dots, \phi_{28}\}, \quad \text{rank } P_0 = 30. \quad (61)$$

The electric vacuum is unique and its full physical gap is exactly three. Six-edge fundamental loops have energy exactly $9/2$ and are excluded from this initial interval. A total spin-degree cutoff is not identified with it.

For $p < q$ put $X_{pq} = W_p W_q$. There are 96 pairs sharing one edge and 310 sharing no edge; 64 of the latter meet only at vertices. On a common edge let E_e be Haar integration, the spin-zero Casimir projection, and split $X_s = E_e X_{pq}$, $X_t = X_{pq} - X_s$. These are the two generated branches of $1/2 \otimes 1/2 = 0 \oplus 1$. Haar averaging commutes with the endpoint gauge actions. With the face orientations retained, the basic conditional identity is

$$\int \chi(AU) \chi(U^{-1}B) dU = \frac{1}{2} \chi(AB). \quad (62)$$

The two exterior three-link paths use disjoint variables. Their conditional Haar norms give $\|X_s\|^2 = 1/64$, $\|X_t\|^2 = 3/64$ and zero cross term. Exact quaternion-polynomial integrations check this result; no Monte Carlo or joint independence of all face holonomies is assumed.

The specified *individually resolved* enrichment consists of these actual functions:

Function	Count	Squared norm	Electric energy
Ω	1	1	0
$\phi_p = 2W_p$	29	1	3
$\chi_p = 4W_p^2 - 1$	29	1	8
$d_{pq} = 4W_p W_q$, edge-disjoint	310	1	6
$s_{pq} = 8X_s$, shared edge	96	1	$9/2$
$t_{pq} = 8X_t$, shared edge	96	3	$13/2$

Every one of the 406 pair parity masks is nonzero, distinct from the others and unequal to a single-face mask. Independent edge-center flips prove the corresponding orthogonalities; spin-one face characters are separately distinguished by their integer representation supports. The two shared branches are orthogonal electric eigenspaces. Thus the actual Gram is diagonal with 465 ones and 96 threes, has zero kernel, and gives quotient rank 561. This is proved from the graph and Haar functions, not assumed from an older example. Smooth electric eigenvectors make the resulting orthogonal P_+ reduce K and its operator and form domains. The span does not fill complete ambient electric shells, and taking only the cyclic closure of summed columns need not separate its individual degenerate channels.

7.3 The complete magnetic matrix and simultaneous outside loading

Assign link-center signs $+1$ on x links, $(-1)^x$ on y links and $(-1)^{x+y}$ on z links. Their product is -1 on every actual face. This Haar-preserving transformation commutes with K and gauge averaging and sends S to $-S$. Vacuum and the quadratic channels are even; the fundamental faces are odd. It proves all same-parity magnetic entries vanish. The pair-mask independence above is a separate argument.

The complete old columns are

$$\begin{aligned} S\Omega &= \frac{1}{2} \sum_p \phi_p, \\ S\phi_p &= \frac{1}{2}\Omega + \frac{1}{2}\chi_p + \frac{1}{2} \sum_{q: e(p) \cap e(q) = \emptyset} d_{pq} + \frac{1}{4} \sum_{q: |e(p) \cap e(q)|=1} (s_{pq} + t_{pq}). \end{aligned} \quad (63)$$

Together with the parity zeros and metric self-adjointness $g_i S_{ij} = g_j S_{ji}$, these determine every retained magnetic entry. In particular a triplet-to-face coefficient is $3/4$, while its face-to-triplet coefficient is $1/4$. Blind coordinate transposition would be wrong. The exact sparse export contains 2,124 directed nonzero entries and a defining zero rule for the other 312,597 entries. The full retained generator is its sum with the listed electric diagonal and $29\lambda I$.

For $A_+ = P_+ LP_+$ and $Q_+ = I - P_+$, define the actual full outside map

$$B = Q_+ LP_+ = -\lambda Q_+ SP_+, \quad BP_0 = 0, \quad \|B\| \leq 29\lambda. \quad (64)$$

K reduction and (63) justify the first two statements. Orthogonal projection in the actual physical Gram and $\|S\| \leq 29$ give the last bound simultaneously for every complex retained input. Every new column and unretained intertwiner is charged. Equivalently, for each basis function b_j its outside column is the full polynomial $-\lambda[Sb_j - \sum_i b_i \langle b_i, Sb_j \rangle / g_i]$. This specifies all columns without claiming an evaluated cubic outside Gram or exact outside norm.

The map is nonzero for $\lambda > 0$: the normalized spin-3/2 same-face character ξ_p has energy 15 and lies outside P_+ , with

$$\langle \xi_p, B\chi_p \rangle = -\lambda/2. \quad (65)$$

Other faces have the wrong center mask for this coefficient. Thus $BP_0 = 0$ does not give autonomous retained dynamics; newly populated channels can load the exterior.

7.4 The full original residual and both nested ground comparisons

Put $m = 29$ and

$$\begin{aligned} w &= \frac{\sqrt{9 + m\lambda^2} - 3}{2}, & h &= \frac{\lambda}{2(3 + w)}, & Z &= 1 + mh^2, \\ f_0 &= \frac{\Omega + h \sum_p \phi_p}{\sqrt{Z}}, & \mu_0 &= m\lambda - w. \end{aligned} \quad (66)$$

These are the exact ground pair of the complete P_0 star matrix. The dark face eigenvalues are $3 + m\lambda$. Positivity, compactness, the electric gap and min-max first establish unique full and compressed grounds with

$$0 \leq \epsilon \leq \mu_+ \leq \mu_0 \leq 29\lambda < 3, \quad E_1(L), E_1(A_+), E_1(A_0) \geq 3. \quad (67)$$

Thus isolation precedes residual inference. Write their rank-one projections as G, G_+, G_0 .

For $F = S^2 - m/4$, complete conditional Haar fourth moments and the pair masks yield

$$\begin{aligned} F &= \frac{1}{4} \sum_p \chi_p + \frac{1}{2} \sum_{\text{disjoint}} d_{pq} + \frac{1}{4} \sum_{\text{shared}} (s_{pq} + t_{pq}), \\ \|F\|^2 &= \frac{m(2m-1)}{16} = \frac{1653}{16}, \quad (L - \mu_0)f_0 = -\frac{2\lambda h}{\sqrt{Z}}F, \\ \rho_0^2 &= \frac{1653\lambda^2 h^2}{4Z}. \end{aligned} \quad (68)$$

The original omitted-channel Gram has diagonal $m/4$, off-diagonal $1/4$, and bright/dark eigenvalues $57/4$ and 7 . It is independently recovered from the full resolved matrix. The residual above is the actual full-Hilbert residual; keeping only the spin-one face terms would miss most of it.

For the uniform coupling cap $\Lambda = 1/100$, define

$$g = 3 - 29\Lambda = 271/100, \quad \bar{r}_0 = 41\Lambda^2/12, \quad \bar{p}_0 = \bar{r}_0/g. \quad (69)$$

The inequalities $h \leq \lambda/6$ and $\sqrt{1653} < 41$ justify the first residual envelope. There are two distinct spectral arguments:

$$\|G - G_0\| \leq \bar{p}_0, \quad \|G_+ - G_0\| \leq \bar{p}_0. \quad (70)$$

The first uses the full spectral measure of f_0 . For the second, $BP_0 = 0$ implies $Lf_0 \in \text{ran } P_+$, so the residual in the separately isolated A_+ spectrum is exactly the same full residual. An internal zero residual of the enriched ground would not replace either argument.

For the exact normalized enriched ground f_+ , $BG_0 = 0$ then gives

$$\rho_+ = \|(L - \mu_+)f_+\| = \|B(I - G_0)f_+\| \leq 29\Lambda\bar{p}_0 =: \bar{r}_+. \quad (71)$$

Its full spectral measure and the nonnegative spectral polynomial $(L - \epsilon)(L - 3)$ give

$$\|G - G_+\| \leq \bar{p}_+ := \bar{r}_+/g, \quad 0 \leq \mu_+ - \epsilon \leq \bar{d}_+ := \bar{r}_+^2/g. \quad (72)$$

Smoothness supplies the second moment needed for the energy estimate. All barred quantities are upper envelopes, not evaluated actual enriched ground data. In exact fractions,

$$\bar{p}_0 = \frac{41}{325200}, \quad \bar{r}_+ = \frac{1189}{32520000}, \quad \bar{p}_+ = \frac{1189}{88129200}, \quad \bar{d}_+ = \frac{1413721}{2865961584000000}. \quad (73)$$

7.5 True output denominator and independently centered heat

Let x be any normalized complex vector in $\text{ran } P_0$ with $\|x - \Omega\| \leq \eta = 1/100$. Following the forward constants, put

$$\begin{aligned} z_0 &= 1 - 29\Lambda^2/72, \quad q_0 = (11/12)\Lambda, \quad b = q_0 + \bar{p}_0 + \eta, \\ D &= z_0 - \eta - \bar{p}_0 = \frac{193136341}{195120000} > 0. \end{aligned} \quad (74)$$

The Ritz overlap and both comparisons in (70) give $\|(I - G)x\|, \|(I - G_+)x\| \leq b$. For the actual full heat,

$$E(\sigma) = e^{-\sigma(L-\epsilon)}, \quad E_+(\sigma) = e^{-\sigma(A_+-\mu_+)}P_+, \quad \|E(\sigma)x\| \geq \|Gx\| \geq D. \quad (75)$$

The denominator is the true full output, not a Ritz norm or a regularizer. The bounds apply to complex preparations and both centered excited gaps are at least g .

Strong differentiation on smooth finite retained vectors and vectorwise integration give the exact identity

$$(E - E_+)(\sigma)x = -\int_0^\sigma E(\sigma - s)[B + (\mu_+ - \epsilon)P_+]E_+(s)x ds. \quad (76)$$

The full E inside the integral retains outside evolution and return. The plus centering defect survives $BP_0 = 0$. No unbounded operator-norm derivative is assumed. Independent ground/excited decompositions give an increasing early bound and decreasing late bound,

$$\begin{aligned} V(\sigma) &= (\bar{r}_+ + \bar{d}_+)\sigma + \frac{29\Lambda b}{g}(1 - e^{-g\sigma}), \\ J(\sigma) &= \bar{p}_+ + (2b + \bar{p}_+)e^{-g\sigma}. \end{aligned} \quad (77)$$

The extra decaying $\bar{p}_+$ term is harmless slack. At the predeclared join $\sigma = 2$, use $1 - e^{-2g} \leq 1$ and a positive rational Taylor sum proving $e^{2g} > 225$. Monotonicity covers $[0, 2]$ and $[2, \infty)$ without a time grid. The forward certificate is

$$\sup_{\sigma \geq 0} \frac{\|(E - E_+)(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{3063228992521}{1418412601938100} < \frac{11}{5000}. \quad (78)$$

The skeptic independently reaches the same rational headline. The reverse proof uses the tighter valid $q_0 = (9/10)\Lambda$ and obtains $3037671524521/1418412601938100 < 43/20000 = 0.00215$. These are conservative exact-semigroup bounds, not measured approximation errors; the larger graph's common certificate is weaker than the earlier graph's sharper AC2 certificate.

At zero time the two exact evolutions agree on this initial class. At zero coupling they agree on retained inputs at every time because P_+ reduces K . Those exact statements supersede positive cap envelopes. Zero-extension has full-input operator norm error one at zero time, so arbitrary full-Hilbert initial accuracy is not claimed. Replacing either actual ground center by 29λ or by an independently rounded energy can produce long-time exponential error and is outside the theorem.

Exact controls test the actual gauge words, every internal face, the strict endpoint, both shared branches, the triplet metric, duplicate-coordinate null relations, individual versus summed channels, new-input spin-3/2 leakage, negative-coupling misuse, complex preparations and center-shift sign. The reverse record preserves an initially nondiscriminating orientation fixture and its discriminating replacements; that control refinement adds no investigation. Exact generator actions are executed, but no numerical heat evaluator or numerical trajectory is supplied. Such an implementation would require an added error budget divided by the same true denominator. The outside cubic Gram, complete ambient-shell enrichment, real-time relative control, volume-uniform limit, canonical/homogeneous matching and continuum Yang–Mills mass gap remain open; scientific priority is unverified.

8 AH2: complete delayed loading at the unchanged physical clock**

AH2 sharpens the early omission estimate on the same 46-link graph and the same rank-561 physical compression. The two independent directions obtain the exact sufficient bound

$$\sup_{\sigma \geq 0} \frac{\|E(\sigma)x - E_P(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{18126701281201}{212761890290715000} \simeq 8.519712462 \times 10^{-5} < 10^{-4}. \quad (79)$$

It holds for every real $0 \leq \lambda \leq 1/100$ and every normalized complex $x \in \text{ran } P_0$ with $\|x - \Omega\| \leq 1/100$. Here both semigroups use their own actual ground energies. This improves AH1's common sufficient ceiling $11/5000$ without narrowing the preparation class, changing the graph or altering the clock. It evaluates no actual heat vector or observed approximation error.

8.1 Full physical domains and all three complementary sectors

Keep precisely (60): the 24 original vertex Gauss actions on all 46 original links, all 29 faces, the invariant product-Haar Hilbert space, and $L = K + \lambda(29 - S)$. Its operator domain is the

invariant product H^2 space, its form domain is invariant H^1 , and smooth invariant Peter–Weyl functions form a core. Bounded real gauge-invariant multiplication preserves these domains and compact resolvent; $0 \leq 29 - S \leq 58$ implies $L \geq K$. The electric vacuum remains unique with full physical gap three. Physical energy and heat time remain $H = \alpha L$ and $\sigma = \alpha t/\hbar$, with all declared reference scales fixed.

Write $P = P_+$ for the admitted enriched projection and set

$$F = P_0 - |\Omega\rangle\langle\Omega|, \quad N = P - P_0, \quad Q = I - P, \quad A = PLP, \quad B = QLP. \quad (80)$$

The dimensions of F and N are 29 and 531. The full outside Q is infinite dimensional. In particular, NLP_0 can be nonzero although $QLP_0 = 0$. Both finite projections reduce K and preserve its operator and form domains. Their ranges consist of smooth physical functions, giving the full block domain $\text{ran } P \oplus QD(K)$ and a legitimate strong-vector Duhamel argument below.

The complete AH1 magnetic action is independently reconstructed from the actual basis, metric adjoints and all old columns: 2,124 nonzero entries and 312,597 structural zeros. The coordinate-dependent link-center transformation makes every new state even and every fundamental face odd. Hence $NSN = 0$ on the whole new block. Its energies are 8 for the 29 spin-one faces, 6 for the 310 disjoint products, $9/2$ for the 96 shared singlets and $13/2$ for the 96 shared triplets; each triplet keeps metric weight three.

Let ϵ, μ be the actual full and retained ground energies, and G, G_P their rank-one projections. The inherited full variational ordering and separate spectral isolation give

$$0 \leq \epsilon \leq \mu \leq \mu_0 \leq 29\lambda < 3, \quad E_1(L), E_1(A) \geq 3. \quad (81)$$

Consequently the complete new diagonal block, not a freely chosen damping term, is

$$D_N = N(A - \mu)N = K_N + (29\lambda - \mu)N \geq dN, \quad d = 9/2. \quad (82)$$

Summing every new row in the physical metric gives

$$(NSF)^*(NSF) = 7I_{29} + \frac{1}{4}J_{29}, \quad C = NAP_0 = -\lambda NSF, \quad C\Omega = 0, \quad \|C\| = \frac{\sqrt{57}}{2}\lambda. \quad (83)$$

For arbitrary complex v , the quadratic form is $7\|v\|^2 + |\sum_p v_p|^2/4$. Its bright eigenvalue is $57/4$ and its 28 dark eigenvalues are seven. All 531 new rows contribute positively to the bright quadratic form; deleting any row changes the exact Gram and its unique largest endpoint. This is an all-input norm proof, not an assumption that the trajectory remains in the bright mode. The complete outside identities are still

$$B = -\lambda QSP = BN, \quad BP_0 = 0, \quad \|B\| \leq 29\lambda. \quad (84)$$

The spin-3/2 same-face witness has coefficient $-\lambda/2$ from the normalized spin-one input, so $B \neq 0$ at positive coupling. The envelope covers all new inputs and all outside channels simultaneously; their exact outside Gram is not evaluated.

8.2 Stationary loading and the complete evolving face source

Put $\Lambda = \eta = 1/100$ and retain the common AH1 spectral envelopes

$$\begin{aligned} g &= 3 - 29\Lambda = 271/100, & r_0 &= 41\Lambda^2/12, & p_0 &= r_0/g, \\ r_P &= 29\Lambda p_0, & p_P &= r_P/g, & \delta_P &= r_P^2/g. \end{aligned} \quad (85)$$

The two separate comparisons $\|G - G_0\| \leq p_0$ and $\|G_P - G_0\| \leq p_0$ are both needed. The first follows from the original Ritz vector's full spectral residual. The second uses the same residual

because $Lf_0 \in \text{ran } P$, with independently established retained isolation. Then $BG_0 = 0$ bounds the enriched full residual by r_P , and the full spectral-angle and Temple estimates give

$$\|G - G_P\| \leq p_P, \quad 0 \leq \mu - \epsilon \leq \delta_P. \quad (86)$$

These symbols denote envelopes, not evaluated projector distances or eigenvectors.

Use the reviewed reverse-AH1 refinement $q_0 = 9\Lambda/10$, with that attribution. It follows from the original Ritz parameter and $\sqrt{29} < 27/5$. Define

$$q = q_0 + p_0 = \frac{14839}{1626000}, \quad b = q + \eta = \frac{31099}{1626000}. \quad (87)$$

The operator inequality $\|FG_P\| \leq \|FG_0\| + \|G_P - G_0\| \leq q$ bounds the actual stationary face component. Both full and retained excited components of every allowed x have norm at most b . Therefore the complete retained evolution satisfies

$$u(\sigma) = e^{-\sigma(A-\mu)}x, \quad \|Fu(\sigma)\| \leq q + be^{-g\sigma}. \quad (88)$$

This spectral decomposition already includes every return through N . The nondecaying q term cannot be discarded: at positive coupling a retained ground with no face component would, by the positive new block and the vacuum-to-face equation, have to be the vacuum, which fails the ground equation. The reverse and skeptic checks also give positive actual-model stationary-face lower witnesses.

For $y = Nu$, the exact coupled equation and its variation-of-constants identity are

$$\begin{aligned} y' &= -D_N y - CFu, & y(0) &= 0, \\ y(\sigma) &= -\int_0^\sigma e^{-(\sigma-s)D_N} CFu(s) ds. \end{aligned} \quad (89)$$

No term is removed from the old-sector equation, including its return term $-C^*Nu$. With $\bar{C} = 19\Lambda/5 > \Lambda\sqrt{57}/2$, equations (82)–(88) imply

$$\|Nu(\sigma)\| \leq Y(\sigma) = \bar{C} \left[\frac{q}{d}(1 - e^{-d\sigma}) + \frac{b}{d-g}(e^{-g\sigma} - e^{-d\sigma}) \right]. \quad (90)$$

Both terms are nonnegative convolutions. They account for stationary and excited loading of all 531 new states. The source is the full u , not a prescribed free face trajectory.

8.3 Actual centers, true denominator and a fixed all-time join

Define $E(\sigma) = e^{-\sigma(L-\epsilon)}$ and $E_P(\sigma) = e^{-\sigma(A-\mu)}P$. Since $u(s)$ stays in a finite smooth full-domain space and $Lu(s)$ is continuous, strong vector differentiation and integration give

$$(E(\sigma) - E_P(\sigma))x = -\int_0^\sigma E(\sigma-s)[B + (\mu - \epsilon)P]u(s) ds. \quad (91)$$

The plus sign of the center defect inside the integrand is essential. The full contraction $E(\sigma-s)$ retains outside propagation and every return path. This is not an unbounded operator-norm differentiation argument.

Using $Bu = BNu$, $\|u\| \leq 1$ and $\bar{B} = 29\Lambda$ yields the increasing early error envelope

$$\begin{aligned} V(\sigma) &= \delta_P \sigma + \bar{B}\bar{C} \left\{ \frac{q}{d} \left[\sigma - \frac{1 - e^{-d\sigma}}{d} \right] \right. \\ &\quad \left. + \frac{b}{d-g} \left[\frac{1 - e^{-g\sigma}}{g} - \frac{1 - e^{-d\sigma}}{d} \right] \right\}. \end{aligned} \quad (92)$$

The leakage part is $\bar{B} \int_0^\sigma Y(s) ds$, so monotonicity and $V(0) = 0$ follow from nonnegative kernels. Integrating the transient convolution over the whole half-line gives $1/(gd)$. Thus a simpler increasing majorant is

$$V(\sigma) \leq \bar{V}(\sigma) = \delta_P \sigma + \bar{B}\bar{C} \left(\frac{q\sigma}{d} + \frac{b}{gd} \right). \quad (93)$$

Its positive intercept is conservative slack; the exact error at zero time is zero.

The original Ritz overlap, full projector comparison and complete complex preparation ball give the genuine output floor

$$\|E(\sigma)x\| \geq \|Gx\| \geq D = 1 - 29\Lambda^2/72 - \eta - p_0 = \frac{193136341}{195120000} > 0. \quad (94)$$

This is the full output, not the retained norm or an arbitrary regularizer. Separately the full and retained spectral tails imply

$$\|(E(\sigma) - E_P(\sigma))x\| \leq J(\sigma) = p_P + (2b + p_P)e^{-g\sigma}, \quad (95)$$

where the extra decaying p_P is harmless slack. At the predeclared join $\sigma = 3$, positive rational Taylor sums prove $e^{3g} > 3000$. The increasing early bound controls $[0, 3]$ and the decreasing late bound controls $[3, \infty)$:

Certificate endpoint	Exact bound
$\bar{V}(3)$	18126701281201/214947118800000000
$J(3)$	$\leq 34696603/1321938000000$
True output floor D	193136341/195120000

The early endpoint dominates. Dividing it by D proves (79). All inequalities use simultaneous cap constants and positive gaps, so they cover the full real coupling interval and all nonnegative real times. The join changes no physical evolution parameter and was not optimized.

The independently frozen skeptic uses the slightly sharper rational choice $\bar{C} = 151\Lambda/40$, giving the separately attributed sufficient fraction

$$\frac{4501862190047}{53190472572678750} \simeq 8.463662706 \times 10^{-5} < 10^{-4}. \quad (96)$$

Its stationary-loading, excited-loading and center-defect absolute contributions at the join are respectively about 6.6605×10^{-5} , 1.7170×10^{-5} and 1.4798×10^{-9} . These are conservative proof budgets, not measured error components; neither certificate is claimed optimal.

8.4 Exact fixtures, discriminating controls and limits

The 27 prescribed records combine $\lambda \in \{0, 1/200, 1/100\}$, $\sigma \in \{0, 1, 3\}$ and three preparations. In addition to Ω , the two nonvacuum states are

$$\sqrt{39999/40000} \Omega + \frac{1}{200} \phi_0, \quad \sqrt{39999/40000} \Omega + \frac{i}{200} \phi_0. \quad (97)$$

They have exact physical norm one and vacuum distance slightly larger than their face amplitude 0.005, but below the theorem radius 0.01. Every record uses the full theorem radius. Rational Taylor and reciprocal enclosures verify scalar certificate functions in (90)–(95); no finite matrix exponential or heat vector is computed. Analytic monotonicity and spectral estimates, not these fixtures, establish continuous and all-time coverage.

Controls check every row's Gram contribution, the triplet metric, the distinction between N and Q , the nonzero outside spin-3/2 channel, stationary loading, actual centers and the fixed

clock. Feedback deletion changes the actual second derivative by C^*CFx . For either allowed nonzero face preparation its squared norm is $(1/200)^2\lambda^4(869/16) > 0$ at positive coupling. The producers preserve their initial nondiscriminating candidates—zero coupling in one direction and vacuum-only second derivative in the other—beside the successful replacements; these are control repairs, not new investigations. A polynomial vanishing at all fixture times explicitly refutes a sampling-only all-time inference.

The forward control also proves $\epsilon < \mu$ at positive coupling using the connected nonpositive-off-diagonal retained matrix, positive retained ground coordinates and its nonzero outside spin-3/2 coefficient. For $x = \Omega$ this gives

$$\frac{d}{d\sigma} [\|E(\sigma)x\|^2 - \|E_P(\sigma)x\|^2] \Big|_{\sigma=0} = 2(\epsilon - \mu) < 0. \quad (98)$$

Thus the exact retained norm can exceed the true output norm and cannot replace (94). This supplies no evaluated enriched ground or quantitative new gap. At zero time, or zero coupling for every time, the exact two heats agree on allowed inputs. Independently rounded centers cannot be exponentiated indefinitely while preserving the theorem.

Both initial freeze inventories omitted the two nested AH1 freeze snapshots. The completed records add exactly those already existing snapshots and the preserved initial draft record. Their reports, checkers, outputs and all initially bound bytes are unchanged; the source inventories and final review verify this disclosed metadata repair. AH2 inherits the reviewed AH1 construction and reverse-origin vacuum-angle refinement; each new checker parses frozen data and independently reconstructs the needed blocks without executing an older checker. No new external primary reading is claimed.

This is one analytic, fixed-graph, same-clock refinement for exact independently centered heat and the whole declared initial ball. It supplies no numerical evaluator, exact outside Gram, arbitrary-full-input or real-time relative theorem, rounded-center robustness, graph-size uniformity, model identification, thermodynamic construction or continuum Yang–Mills result. The remaining early bound is dominated by stationary loading. A smaller sufficient fraction is not an observed improvement in an actual computed trajectory.

9 AJ1: the actual homogeneous physical GNS representation**

AJ1 proves local regularity and physical cyclic-space identification for the actual I1 positive-orthant homogeneous omitted-coupling state. The new estimate is a local reference-energy bound,

$$\text{Tr}(\rho_{\Lambda,R} H_{0,R}) \leq 2M n_{\Lambda}(R) \leq 8M |R| = 56|\tau||R|, \quad M = 7|\tau|. \quad (99)$$

Finite-region compact resolvent turns this estimate into local trace-norm convergence and normality of the represented local algebras. These properties justify continuous endpoint gauge implementation and compatible Haar averaging. The full bounded local invariant algebra generates exactly the joint fixed-vector space, which reduces the actual closed, centered I1 generator. The spectral threshold $\alpha/16$ and symbolic small-coupling interval are inherited; no nonzero physical excitation or numerical nonzero coupling interval is constructed here.

9.1 The complete I1 model and the inherited limit topology

A coarse site $b = (i, j, k) \in \mathbb{Z}_+^3$ owns every positive-direction original link whose tail lies in

$$T_b = \{(4i + r, 2j + s, k) : 0 \leq r < 4, 0 \leq s < 2\}. \quad (100)$$

Euclidean division gives unique ownership. Its 24 Haar link factors contain the complete ten-link selected three-face strip and fourteen free links. Link heads can lie outside this tail-owned region

and retain their original gauge actions. The tensor factors $\mathcal{K}_b = L^2(\text{SU}(2)^{24})$ are the unreduced spaces; a tensor factorization of the Gauss-constrained space is not assumed.

The fixed selected strip coefficients obey $|\lambda_L|, |\lambda_R| \leq \alpha/2$ and $|\mu| \leq \alpha/8$. With $\delta = \alpha/8$, let

$$h_b = \delta^{-1} \left(H_{\text{strip},b} - E_{\text{strip},b} + \alpha \sum_{e \text{ mfree at } b} C_e \right) \geq I - P_b, \quad h_b \Omega_b^0 = 0. \quad (101)$$

The onsite vacuum is unique. On a finite coarse cuboid Λ , the actual normalized Hamiltonian is

$$\begin{aligned} \hat{H}_\Lambda &= H_{0,\Lambda} + \sum_{b \in S \subset \Lambda} \phi_b, & H_{0,\Lambda} &= \sum_{b \in \Lambda} h_b, \\ S &= \{0, e_x, e_y, e_z\}, & \phi_b &= -\frac{\tau}{3} \sum_{f \in O_b} W_f, \quad |O_b| = 21, \quad \|\phi_b\| = M. \end{aligned} \quad (102)$$

All 21 omitted faces are retained only when their *whole* four-site star is contained. Their actual two- and three-site support multiplicities are 1, 3, 1, 10, 2, 4 for $\{0, e_x\}$, $\{0, e_y\}$, $\{0, e_x, e_y\}$, $\{0, e_z\}$, $\{0, e_x, e_z\}$ and $\{0, e_y, e_z\}$. The upper norm bound uses $\|W_f\| = 1$; positive-Haar-measure neighborhoods of the identity attain its essential supremum. Thus a finite same-sign sum of N complete anchors has norm NM , without a bounded global homogeneous perturbation as $N \rightarrow \infty$.

Only the inherited range

$$|\tau| < \tau_* := \frac{\min\{c_1(S), 1/[2c_2(S)]\}}{7} \quad (103)$$

is used. Its positive source constants remain unevaluated. It already supplies unique finite grounds Ω_Λ , the specified whole-star orthant state ω , its GNS triple $(\pi, \mathcal{H}_\omega, \Omega)$, and a nonnegative centered self-adjoint G with unique ground and spectrum in $\{0\} \cup [1/2, \infty)$. The source topology consists of convergence of each bounded local expectation and tested resolvent matrix elements:

$$\begin{aligned} \langle \Omega_\Lambda, A \Omega_\Lambda \rangle &\longrightarrow \omega(A), & A &\in \bigcup_{R \text{ mfinite}} B(\mathcal{K}_R), \\ \langle R_\Lambda(z) A \Omega_\Lambda, B \Omega_\Lambda \rangle &\longrightarrow \langle (G - z)^{-1} \pi(A) \Omega, \pi(B) \Omega \rangle, \\ R_\Lambda(z) &= (\hat{H}_\Lambda - E_\Lambda - z)^{-1}, & z &\notin \mathbb{R}. \end{aligned} \quad (104)$$

Here A, B are fixed bounded local operators. This is not a same-Hilbert-space strong resolvent limit in J2's summable representation.

The orthant transfer keeps independent reference spectators at negative sites of $\mathbb{Z}^3$ and sets outside anchor interactions to zero. Nonnegative star offsets prevent coupling to those spectators. Finite grounds factor, and finite centered resolvents on spectator-vacuum vectors equal the orthant resolvents. The local-normality argument below supplies factorization of the actual local limiting state and density of represented product excitation vectors, including for the full bounded local factor algebra. Vanishing off-sector matrix elements and the tested limit then identify a reducing orthant restriction. This is the specified boundary prescription; equality with every other thermodynamic boundary limit is not claimed.

Yarotsky's source creation core uses $u_I \in \mathcal{H}'_I \cap D(H_{0,I})$ and $\hat{u}_I = |u_I\rangle \langle \Omega_I^0|$. The domain condition makes $[H_0, \hat{u}_I] = |H_{0,I} u_I\rangle \langle \Omega_I^0|$ bounded, while only finitely many bounded star commutators meet I . Density and essential self-adjointness are inherited from the source expansion, not re-proved here. An arbitrary bounded local A need not put $\pi(A)\Omega$ in the operator or form domain of G . The producer proofs below use the tested closed-resolvent limit to avoid that shortcut; the skeptic separately checks covariance on the domain-qualified source core and its closure.

9.2 A domain-valid local reset and complete incident-star counting

The 24-link electric sum has compact resolvent: below a fixed energy only finitely many Peter–Weyl labels occur, each with finite multiplicity. Bounded selected Wilson terms and the exact ground

subtraction preserve its full product H^2 operator domain, H^1 form domain, and compact resolvent. The same holds for each finite-region sum $H_{0,R}$.

The reverse and skeptic additionally use the strip potential bound $9\alpha/8$ and $-9\alpha/8 \leq E_{\text{strip},b} \leq 0$ to obtain

$$\left\| h_b - 8 \sum_e C_e \right\| \leq 18, \quad h_b \geq 8 \sum_e C_e - 9. \quad (105)$$

For $Q_{R,T} = \mathbf{1}_{[0,T]}(H_{0,R})$, the reverse's explicit counting bound is

$$\text{rank } Q_{R,T} \leq \left[\sum_{k=0}^{\lfloor 4B_R/3 \rfloor} (k+1)^2 \right]^{24|R|}, \quad B_R = (T + 9|R|)/8. \quad (106)$$

It uses $k(k+2) \geq 3k$ for integer $k = 2j \geq 0$. This is a loose spectral-counting comparison, not equality of interacting and bare-electric spectral projections or an evaluated interacting spectrum.

Let $\rho_\Lambda = |\Omega_\Lambda\rangle\langle\Omega_\Lambda|$ and reset only a finite region $R \subset \Lambda$ to its reference vacuum:

$$\rho'_\Lambda = |\Omega_R^0\rangle\langle\Omega_R^0| \otimes \text{Tr}_R \rho_\Lambda, \quad \Omega_R^0 = \bigotimes_{b \in R} \Omega_b^0. \quad (107)$$

The original finite ground has finite reference energy by bounded perturbation. For the reset density, bounded positive spectral truncations and the partial-trace identity preserve exterior reference energy; monotone convergence then justifies the unbounded statement. Its regional reference energy is zero. The reset therefore has finite form energy and is an admissible mixed trial density. It is unnecessary to place every vector in that density's support in the operator domain.

All relevant incident anchors are

$$I_\Lambda(R) = \{b : b + S \subset \Lambda, (b + S) \cap R \neq \emptyset\}, \quad n_\Lambda(R) = |I_\Lambda(R)|. \quad (108)$$

Every anchor is $r - s$ for some $r \in R, s \in S$. Thus, including incoming and outgoing stars,

$$n_\Lambda(R) \leq n_+(R) := |(R - S) \cap \mathbb{Z}_+^3| \leq 4|R|. \quad (109)$$

Apply the finite variational inequality to (107). Exact scalar energies, exterior reference terms and disjoint-star expectations cancel:

$$\begin{aligned} 0 &\leq \text{Tr}(\rho'_\Lambda \hat{H}_\Lambda) - E_\Lambda \\ &= -\text{Tr}(\rho_\Lambda H_{0,R}) + \sum_{b \in I_\Lambda(R)} \text{Tr}[(\rho'_\Lambda - \rho_\Lambda) \phi_b]. \end{aligned} \quad (110)$$

Each difference costs at most $2M$ in absolute value, proving (99) for every finite region and volume. The sum on the right is nonnegative for the actual ground; positivity of each interaction difference is not assumed. The reverse retains the sharper regional constant $C_R = 2Mn_+(R)$, while the common forward/skeptic headline uses $C_R \leq 8M|R|$. No extensive global bound or numerical source threshold is substituted.

9.3 Local trace-norm convergence and normality of the representation

For $T > 0$, the finite-rank cutoffs $Q = Q_{R,T}$ increase strongly to the identity. Spectral order gives

$$\text{Tr}[\rho_{\Lambda,R}(I - Q)] \leq C_R/T =: \varepsilon_T. \quad (111)$$

Finiteness of C_R , rather than its smallness, is enough. For a trace-one density ρ with $e = \text{Tr}[\rho(I - Q)]$, the forward and skeptic use

$$\|\rho - Q\rho Q\|_1 \leq 2\sqrt{e}. \quad (112)$$

Indeed $\rho - Q\rho Q = (I - Q)\rho + Q\rho(I - Q)$ and Hilbert–Schmidt factorization bounds both trace norms by $\sqrt{e}$. The reverse separately proves the valid slightly looser $2\sqrt{e} + e$ by splitting the two off-diagonal blocks and omitted positive diagonal. These different estimates lead to the same local conclusion.

For each fixed finite-rank Q , the matrix entries of $Q\rho_{\Lambda,R}Q$ converge by the inherited bounded-local state limit. Uniform tails make the densities trace-norm precompact, and every cluster density represents that same limiting functional. Hence

$$\|\rho_{\Lambda,R} - \rho_R\|_1 \longrightarrow 0, \quad \omega(A) = \text{Tr}(\rho_R A), \quad \text{Tr}(\rho_R H_{0,R}) \leq C_R. \quad (113)$$

Equivalently the forward finite-block argument gives a limiting trace-distance upper bound $4\sqrt{\varepsilon_T}$ before $T \rightarrow \infty$. This concerns every fixed finite region, including unions of supports. It supplies no volume rate or global infinite-volume trace-norm convergence. At $\tau = 0$ the reset estimate forces the local reference vacuum.

Normality of the actual state is not yet the whole averaging premise. For bounded local B in a region K , the positive vector functional

$$A \in B(\mathcal{K}_R) \longmapsto \langle \pi(B)\Omega, \pi(A)\pi(B)\Omega \rangle = \omega_{R \cup K}(B^*AB) \quad (114)$$

is normal by (113) on the union region. Such cyclic vectors are dense. Their vector functionals approximate those of arbitrary vectors in norm; normal functionals form a norm-closed predual. Polarization therefore proves that π is normal on each original local factor. In particular, bounded local strong-* limits and the required local weak-operator averages can be tested through this representation. No normality on a global infinite tensor-product von Neumann algebra is claimed.

The final skeptical review makes the spectator-extension density step explicit. The same reset and compactness proof applies there, with zero interaction cost for spectator sites. The normal finite-region limit is the positive-region density tensored with the pure spectator vacuum. Finite-rank spectator projections converge boundedly strongly to the identity. Normality therefore allows any bounded local operator applied to that vacuum to be approximated in GNS vector norm by finitely many spectator columns, each a bounded positive-region operator times a finite-rank spectator creation. These represented product excitation vectors are dense even though the algebraic tensor product is not asserted norm dense in $B(\mathcal{K}_+ \otimes \mathcal{K}_-)$. There is no use of physical cyclicity in this step.

Projection onto the spectator vacuum preserves the source’s domain-qualified creation core: spectator excitations are killed, while positive-region creations survive. It commutes with the decoupled reference and retained interaction there. Essential self-adjointness and graph closure give a reducing orthant restriction and its projected core; equivalently, off-sector tested resolvent elements vanish on the dense product vectors just justified. This completes the source-correct orthant transfer used in (104).

9.4 Every endpoint, continuous gauge implementation and compatible averages

An original link has the full endpoint action $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$. Finite products of left/right translations are strongly continuous on the original link spaces, first on the Peter–Weyl span and then by density. They preserve every Casimir domain, selected face and omitted closed Wilson word, so they commute with each finite Hamiltonian. All external head endpoints are retained.

Uniqueness makes a finite ground transform by a continuous one-dimensional character of a finite product of $\text{SU}(2)$. Such a character is trivial: an $\text{SU}(2)$ element is conjugate to its inverse, and connectedness forces the resulting value in $\{1, -1\}$ to equal one. Thus the finite ground vectors and onsite vacua are fixed. Passing their expectations to the limit proves invariance of ω under each local automorphism β_g . The prescription

$$U_\omega(g)\pi(A)\Omega = \pi(\beta_g(A))\Omega \quad (115)$$

respects the GNS null ideal, preserves inner products, has inverse g^{-1} and fixes Ω . For local A , its original orbit is bounded and strong-* continuous within the same owned-link factor. Local normality yields

$$\| [U_\omega(g) - U_\omega(g_0)]\pi(A)\Omega \|^2 = \omega((\beta_g A - \beta_{g_0} A)^*(\beta_g A - \beta_{g_0} A)) \longrightarrow 0. \quad (116)$$

Density extends this to the whole GNS space for every finite endpoint group. This is strong unitary continuity, not point-norm continuity of conjugation on every bounded local operator.

Let $V(R)$ contain all endpoints of the $24|R|$ owned links. In the original local factor define the weak-operator Haar average

$$\mathbb{E}_R(A) = \int_{\text{SU}(2)^{V(R)}} \beta_g(A) dg. \quad (117)$$

It is a unital positive contraction, stays in $B(\mathcal{K}_R)$ and is invariant under all local gauge transformations: outside endpoints act trivially. The averages are compatible with support enlargement. To compare it with the represented vector average, test against any $\pi(B)\Omega$. The normal functional $X \mapsto \omega(B^*X)$ on the union region passes through the local weak-operator integral. Density and strong continuity then give

$$\pi(\mathbb{E}_R(A))\Omega = \int_{\text{SU}(2)^{V(R)}} U_\omega(g)\pi(A)\Omega dg. \quad (118)$$

This proves the compatibility actually needed. An arbitrary GNS representation would not justify the interchange.

9.5 The physical null quotient and reducing closed generator

Let $\mathcal{A}_{\text{phys}}$ be the norm closure of the *full bounded local invariant algebra*, and define

$$\begin{aligned} \mathcal{H}_{\text{cyc}} &= \overline{\pi(\mathcal{A}_{\text{phys}})\Omega}, \\ \mathcal{H}_{\text{fix}} &= \{\xi : U_\omega(g)\xi = \xi \text{ for every finite local } g\}. \end{aligned} \quad (119)$$

Equation (115) gives the first inclusion. Conversely, approximate a fixed ξ by a local $\pi(A)\Omega$. The finite endpoint Haar projection fixes ξ and contracts the error, so (118) gives

$$\|\pi(\mathbb{E}_R(A))\Omega - \xi\| \leq \|\pi(A)\Omega - \xi\|, \quad \mathcal{H}_{\text{cyc}} = \mathcal{H}_{\text{fix}}. \quad (120)$$

Each approximant uses only a finite compact endpoint product. No infinite-group Haar construction or Wilson-only density premise enters. For $\mathcal{N}_{\text{phys}} = \{A \in \mathcal{A}_{\text{phys}} : \omega(A^*A) = 0\}$, the map $[A] \mapsto \pi(A)\Omega$ is an isometry of the physical GNS quotient, with inner product $\omega(A^*B)$, onto a dense subspace of (120). Its completion is the desired intertwining unitary; state faithfulness is unnecessary.

Finite centered resolvents commute with the original endpoint actions and their grounds are fixed. Applying (104) to $\beta_g(A), \beta_g(B)$ and then using density gives the actual bounded-resolvent identity

$$U_\omega(g)^*(G - z)^{-1}U_\omega(g) = (G - z)^{-1}, \quad z \notin \mathbb{R}. \quad (121)$$

Both the resolvent and its adjoint preserve the joint fixed space. Hence that space reduces every spectral projection of the closed G , not merely a formal local commutator. The physical generator has the exact domains

$$\begin{aligned} G_{\text{phys}} &= G|_{\mathcal{H}_{\text{cyc}}}, \\ D(G_{\text{phys}}) &= D(G) \cap \mathcal{H}_{\text{cyc}}, \end{aligned} \quad D(G_{\text{phys}}^{1/2}) = D(G^{1/2}) \cap \mathcal{H}_{\text{cyc}}. \quad (122)$$

It is a self-adjoint restriction with the same unique centered vacuum. In fixed physical units,

$$H_{\text{phys}} = \frac{\alpha}{8} G_{\text{phys}}, \quad \text{spec}(H_{\text{phys}}) \subset \{0\} \cup [\alpha/16, \infty), \quad \text{frequency generator} = H_{\text{phys}}/\hbar. \quad (123)$$

The inherited form inequality restricts to the stated form domain. A vacuum-only physical space would satisfy this spectral exclusion as well, so the argument does not yet prove nonzero homogeneous Wilson variance or even a nonzero physical excitation. The threshold is inherited from I1, not a measured mass or a new numerical stability interval.

9.6 Three complete geometry fixtures, provenance and limits

The independently written checks enumerate original links, all endpoints, complete selected/free splits, every omitted face word and full-star retention. The declared cuboids give:

Coarse sides	Sites	Links	Endpoints	Stars	Omitted faces
(2, 2, 2)	8	192	120	1	21
(3, 2, 2)	12	288	176	2	42
(3, 3, 3)	27	648	342	8	168

The boundary singleton, adjacent pair and interior singleton have orthant incidence counts 1, 2, 4, giving reverse regional coefficients $C_R/M = 2, 4, 8$. In the two smaller fixtures the nominal interior singleton has no retained incident star; the all-volume count is not silently substituted for that finite-boundary value. In the largest fixture it receives three incoming stars as well as its own. Whole-star counting charges 84 group faces there, although only 49 individual faces actually touch the singleton. Individually contained omitted-face counts in the three cuboids are instead 70, 110, 336, a different boundary Hamiltonian.

Rational quaternion tests retain all head and tail actions. Finite Q_8 averages verify specified fundamental first moments and closed traces, not full $SU(2)$ Haar averaging of arbitrary operators. Entangled reset fixtures check energy cancellation; the all-volume variational and unbounded-form argument proves (99). Separate analytic countermodels show that weak-star limits of normal states can be singular when energy escapes; strong unitary continuity need not give point-norm conjugation continuity; a weak-operator average need not pass through an arbitrary representation; a bounded rank-one local operator can create a vector outside the generator domain; and a nonreducing compression cannot replace (121). Their finite diagnostics do not prove the infinite statements by sampling.

The reverse preserves a missing-head control whose first gauge element was the identity, together with its discriminating replacement. Its later administrative portability repair archives six original files and changes only the checker's source-binding function to use owned instruction snapshots. The mathematical report and geometry bytes are unchanged; result fields outside binding metadata and all scientific checker logic are unchanged. The active freeze discloses post-exchange access and the earlier independently frozen derivation. This is an administrative repair, not a new independent proof or investigation.

The reset route was an advisor-suggested prospective method shared before the contract; the three derivations independently establish its application. They inherit the external cluster-expansion theorem and its specified core, not a fresh full proof audit. AJ1 supplies local regularity, compatible gauge averaging and the actual homogeneous physical restriction. It establishes no selected positive numerical τ , volume convergence rate, Wilson-only algebra equality, nonzero Wilson fluctuation, boundary-state equivalence, bounded global dressing unitary, summable-model identification, continuum construction or new mass-gap solution.

10 AJ2: a nonzero Wilson fluctuation in the homogeneous state**

AJ2 proves that one specified original Wilson multiplication observable has strictly positive variance in the actual I1/AJ1 homogeneous state. Its centered vector is a nonzero physical vector orthogonal to the vacuum, so AJ1's positive spectral sector is now shown to be nonempty. For every finite physical imaginary time,

$$0 < C(t) \leq v \exp\left(-\frac{\alpha t}{16\hbar}\right), \quad 0 \leq t < \infty, \quad v = \omega(W^2) - \omega(W)^2 > 0. \quad (124)$$

The variance is not numerically evaluated. The proof uses the actual multiplier's Haar-null point levels and AJ1's normal local density, without assuming faithfulness or an interacting Haar law. The decay ceiling uses the inherited physical spectral exclusion; it identifies no mass or threshold eigenvalue.

10.1 The actual face, complete factors and all endpoint actions

Keep the same positive-orthant homogeneous omitted interaction (102), fixed selected-strip reference, whole-star finite boundaries, and symbolic range (103). Zero and either sign of τ are included; no chosen positive numerical coupling is certified. Every site still owns 24 original links, and every retained anchor carries all 21 omitted faces. AJ1 supplies the actual normal local densities, full physical bounded-local algebra, physical cyclic/fixed identification and closed centered restriction

$$H_{\text{phys}} = \frac{\alpha}{8} G_{\text{phys}}, \quad \ker H_{\text{phys}} = \mathbb{C}\Omega, \quad \text{spec}(H_{\text{phys}}) \subset \{0\} \cup [\Delta, \infty), \quad \Delta = \alpha/16. \quad (125)$$

These premises refer to the actual homogeneous state, not the dyadic state or a finite AH graph. The spacing, α , E_* and $\hbar$ retain their declared positive physical meanings.

Let $o = (0, 0, 0)$, $x = (1, 0, 0)$ and $z = (0, 0, 1)$ denote original lattice vertices. The ordered omitted xz face is $o \rightarrow x \rightarrow x + z \rightarrow z \rightarrow o$, with

$$U_{\partial} = U_x(o)U_z(x)U_x(z)^{-1}U_z(o)^{-1}, \quad W = \frac{1}{2} \text{Tr } U_{\partial}. \quad (126)$$

Its four distinct stored positive links have respective coarse owners o, o, z, o under the tail map $(a_x, a_y, a_z) \mapsto (\lfloor a_x/4 \rfloor, \lfloor a_y/2 \rfloor, a_z)$. Thus its minimal complete factor region is $R = \{o, z\}$, containing **48 original links**, twenty selected-strip links and twenty-eight free links. The 16 tails have coordinates $0 \leq i \leq 3$, $0 \leq j \leq 1$, $0 \leq k \leq 1$. Add four x heads at $(4, j, k)$, eight y heads at $(i, 2, k)$ and eight z heads at $(i, j, 2)$ to obtain all **36 endpoint vertices**.

The four loop vertices suffice to see this particular trace's cancellation, but are not the endpoint set for an arbitrary operator on the complete local factor. Under every original action $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$, the ordered word telescopes to $g_o U_{\partial} g_o^{-1}$. Therefore W is in the full bounded local physical algebra. It is a real self-adjoint multiplier, with $|W| \leq 1$. The two-site observable support is distinct from the four-site Hamiltonian anchor star; no two-site Hamiltonian with that whole star retained is inferred.

10.2 Conditional Haar integration proves every point level is null

Work on the original unreduced local multiplication space

$$\mathcal{K}_R = L^2(X, \mu_H), \quad X = \text{SU}(2)^{48}, \quad (127)$$

with normalized product Haar measure. Fix the other 47 original link variables. The reverse and skeptic integrate $U_x(o)$, for which $U_{\partial} = U_x(o)A$ with fixed $A \in \text{SU}(2)$. Right translation

preserves Haar. The forward instead integrates $U_z(o)$, giving $U_\partial = AU_z(o)^{-1}$; inversion and left translation give the same conclusion. These are independent original-link coordinates, not independent plaquette holonomies. The other 44 non-face links are normalized spectators.

For a Haar unit quaternion, the half trace is its scalar coordinate $s = \cos \theta$. The normalized angular density $(2/\pi) \sin^2 \theta d\theta$ on $0 \leq \theta \leq \pi$ yields

$$d\nu_H(s) = \frac{2}{\pi} \sqrt{1-s^2} \mathbf{1}_{[-1,1]}(s) ds. \quad (128)$$

Its integral is one. Every singleton has measure zero, including zero and both endpoints; outside $[-1, 1]$ the level is empty. Conditional integration and Fubini give, for every Borel set B and every real c ,

$$\mu_H\{W \in B\} = \nu_H(B), \quad \mu_H\{W = c\} = 0. \quad (129)$$

The original multiplication spectral projection is $E_W(B) = M_{\mathbf{1}_{\{W \in B\}}}$. It is zero exactly when $\nu_H(B) = 0$, hence

$$E_W(\{c\}) = 0, \quad \ker(W - cI) = \{0\} \quad (c \in \mathbb{R}). \quad (130)$$

Every interior open interval has positive Haar measure. Consequently the multiplier has spectrum $[-1, 1]$, norm one and no point eigenprojection, including at its endpoints. These are original-operator statements. They do not say that the interacting state has the Haar probability law or assigns positive weight to every such interval.

10.3 Strict actual variance without faithfulness

AJ1 supplies a positive trace-class density ρ_R of trace one for the actual restriction of ω to $B(\mathcal{K}_R)$. Put $m = \omega(W) \in \mathbb{R}$. Boundedness makes the moments finite, and

$$v = \text{Tr}[\rho_R(W - mI)^2] = \|(W - mI)\rho_R^{1/2}\|_{\text{HS}}^2. \quad (131)$$

If $v = 0$, the range of $\rho_R^{1/2}$ would lie in the zero kernel (130). This would force $\rho_R = 0$, contradicting its trace. Thus $v > 0$ for every actual state throughout the inherited symbolic interval, without narrowing that interval by a perturbative continuity estimate.

An equivalent configuration argument writes $\rho_R = \sum_j p_j |\psi_j\rangle\langle\psi_j|$. Tonelli gives $r(q) = \sum_j p_j |\psi_j(q)|^2 \geq 0$ with $\int r d\mu_H = 1$, and

$$v = \int_X (W(q) - m)^2 r(q) d\mu_H(q). \quad (132)$$

Zero variance would concentrate all this integrable mass on a Haar-null level. This construction uses the density's spectral decomposition, not an assumed pointwise diagonal kernel. The density may be rank one or vanish on large subspaces. Neither faithfulness, global trace-class representability nor an interacting density bounded below is required. The actual Wilson distribution is absolutely continuous with respect to (128); converse absolute continuity and full support are not asserted.

Since $W - mI$ is bounded local and invariant, its physical GNS class is nonzero:

$$\chi = (\pi(W) - mI)\Omega \in \mathcal{H}_{\text{cyc}}, \quad \langle \Omega, \chi \rangle = 0, \quad \|\chi\|^2 = v > 0. \quad (133)$$

It is precisely the image of $[W - mI]$ under AJ1's quotient isometry. The physical space therefore contains the vacuum and a nonzero orthogonal vector. No operator-domain assertion is needed for this nonvacuity statement.

10.4 A nonzero physical spectral measure and positive heat correlation

For the actual self-adjoint restriction (125), define

$$\nu_\chi(B) = \langle \chi, E_{H_{\text{phys}}}(B)\chi \rangle. \quad (134)$$

This positive finite measure has total mass v , no atom at zero by ground uniqueness and centering, and support in $[\Delta, \infty)$. In particular the actual positive-energy physical sector is nonempty. At every finite real $t \geq 0$, bounded spectral calculus gives

$$C(t) = \langle \chi, e^{-tH_{\text{phys}}/\hbar} \chi \rangle = \int_{[\Delta, \infty)} e^{-tE/\hbar} d\nu_\chi(E). \quad (135)$$

Some finite interval $[\Delta, L]$ has positive measure, so its strictly positive integrand proves $C(t) > 0$. On the entire support the integrand is at most $e^{-t\Delta/\hbar}$, proving (124). Also $C(0) = v$, bounded dominated convergence gives continuity at zero, and the upper bound gives $C(t) \rightarrow 0$ at large imaginary time. No universal lower decay estimate follows.

The actual mean is retained. The uncentered correlation would be

$$\langle \pi(W)\Omega, e^{-tH_{\text{phys}}/\hbar} \pi(W)\Omega \rangle = m^2 + C(t). \quad (136)$$

There is no assumption that the interacting m vanishes. The frequency scale is $\alpha/(16\hbar)$; the dimensionless normalized threshold $1/2$ cannot be inserted as a physical frequency.

The bounded semigroup is defined on every Hilbert vector, so neither $\chi \in D(H_{\text{phys}})$ nor $\chi \in D(H_{\text{phys}}^{1/2})$ is used or proved. Those would require respectively finite second or first energy moments. No derivative at zero is inferred from locality. The support lower bound need not be attained, carry an atom at $\alpha/16$, or equal the lowest overlapping energy. Real-time phase evolution gives no magnitude-decay theorem. AJ2 identifies a nonzero positive measure, not its energies, a normalizable excited eigenvector, a sharp asymptotic rate or a measured mass.

10.5 Reference diagnostics, countermodels and scope

The reference Haar moments

$$\mathbb{E}_H W = 0, \quad \mathbb{E}_H W^2 = 1/4, \quad \mathbb{E}_H W^4 = 1/8 \quad (137)$$

check normalization and conditional integration. They are not assigned to the interacting state. A rank-one Haar density is normal but nonfaithful and still has positive variance. Another normalized normal vector, $2W$, instead has variance $1/2$. Thus normality does not fix the moment values.

More strongly, the normal rank-one states supported in $|W| < \varepsilon$ have positive variance at most ε^2 , while the band has positive Haar measure. Their variances can approach zero. A weak-star cluster subnet can have zero variance and must be singular by (131). This disproves a positive uniform variance floor over *all normal states*; it is not a failure theorem for a possible future quantitative bound on the particular ground-state family. Nonconstancy alone also fails: a step multiplier has a positive-measure constant region and can have zero variance in a normal state supported there. Haar-null levels are the decisive premise.

Other explicit countermodels separate a vacuum-only restricted gap from a nonzero excitation, an upper heat ceiling from a lower decay law, a support endpoint from a threshold atom, and centered heat from a vacuum plateau. A positive-energy atom has constant real-time correlation modulus. Bounded self-adjoint rank-two observables can create vectors with divergent form or operator energy moments; those abstract examples do not assert divergence for the actual Wilson vector.

The exact checkers reconstruct all 48 links, 36 endpoints, coarse owners and the oriented four-link word. Rational quaternion transformations verify full head/tail covariance and charged or missing-head controls. They also check the conditional coefficient normalization and reference moments, with all attempted controls retained. These finite diagnostics do not prove the Haar-null or infinite-state claims by sampling; conditional integration, trace-class normality, the GNS quotient and the spectral theorem supply those arguments.

The advisor supplied the observable and proposed the level-set route before the three independent derivations. I1/AJ1, the immutable seven-loop draft and network are inherited premises; no new external theorem or historical reading is claimed. The original independent skeptic froze its derivation before exchange and subsequently reviewed both producers and replayed their outputs. After a reviewer handoff, another model-agent reviewer read that preserved evidence and independently audited the final admission specification and source closure. The closeout did not replace the original derivation or producer replays, add another pre-exchange proof, or execute another research loop.

This investigation proves qualitative nonvacuity and (124) in the actual specified homogeneous state. It supplies no numerical variance margin, evaluated stability radius, energy-moment result, global faithfulness, Wilson-only algebra completion, alternative boundary-state equality, physical model matching, lattice-spacing limit or continuum Yang–Mills construction. Scientific priority and a percentage of continuum-theorem completion are not established.

11 AK1: a quantitative actual Wilson variance floor**

For the same original origin xz Wilson multiplier (126) in the actual I1/AJ1/AJ2 homogeneous state, both independent producers prove

$$\text{Var}_\omega(W) \geq \frac{119}{576} > \frac{1}{5}, \quad |\tau| < \tau_*, \quad |\tau| \leq 2^{-16}. \quad (138)$$

The independently frozen skeptic proof gives the stronger, separately attributed lower bound 131/576 on the same regime. These are sufficient dimensionless signal bounds, not evaluated interacting variances or optimal constants. The additional numerical cap does not evaluate the positive source constants in $\tau_* = \min\{c_1(S), 1/[2c_2(S)]\}/7$, or certify that any chosen positive numerical coupling lies below it. AJ2's qualitative positivity on the larger symbolic interval remains unchanged.

Keep $|\lambda_L|, |\lambda_R| \leq \alpha/2$ and $|\mu| \leq \alpha/8$ fixed in their inherited ranges. Both bounds are uniform over those choices, with original spacing, α , E_* and $\hbar$ positive and fixed. The actual local density is generally mixed. The energy estimate used below concerns its normalized *reference* energy, not the physical spectral first moment of a Wilson-excited vector.

11.1 Complete region, incident stars and the actual energy premise

The complete unreduced region remains $R = \{0, e_z\}$, with 48 original links, twenty selected-strip and twenty-eight free links, and all 36 endpoint groups. Its four oriented Wilson edges retain signs $+, +, -, -$ and owners $0, 0, e_z, 0$. Full endpoint transformations telescope to conjugation at the origin, so W is bounded, self-adjoint and physical, with $\|W\| = 1$. Neither the four loop vertices nor a tensor product of constrained physical factors replaces this region.

For the inherited $S = \{0, e_x, e_y, e_z\}$, each complete Hamiltonian group has all 21 omitted faces, norm $M = 7|\tau|$, and is retained only when $b + S \subset \Lambda$. An incident anchor has $r = b + s$ for some $r \in R, s \in S$. Conversely every such anchor has a star meeting R . Hence

$$\mathcal{I}_+(R) = (R - S) \cap \mathbb{Z}_+^3 = \{0, e_z\}, \quad \mathcal{I}_\Lambda(R) = \{b \in \mathcal{I}_+(R) : b + S \subset \Lambda\}. \quad (139)$$

The group at zero meets both sites and is counted once. The two-factor observable cover contains no whole star, although ten individual omitted faces fit within it; a two-factor boundary Hamiltonian is not substituted for the actual environment.

The independent reconstructions give the following boundary-complete counts. Charged faces are all faces in incident groups, with group multiplicity.

Coarse cuboid	Links	Endpoints	All stars	Incident stars	Charged faces
(2, 2, 2)	192	120	1	1	21
(3, 3, 3)	648	342	8	2	42
(4, 4, 4)	1536	736	27	2	42

For a cube with side count n , the global counts follow from $8n^3$ original tails, three links per tail, $14n^2$ added heads and $(n-1)^3$ retained stars. More generally (139) proves at most two incident stars for every allowed containing cuboid; the finite table is not the argument for that universal bound.

Retain the exact strip ground scalar and all free-link energies:

$$h_b = \frac{H_{\text{strip},b} - E_{\text{strip},b} + \alpha \sum_e \text{free } C_e}{\delta}, \quad \delta = \alpha/8, \quad h_b \geq I - P_b, \quad (140)$$

where P_b is the unique reference-vacuum projector. The strip gap is at least δ , and a free rotor's gap $3\alpha/4$ equals 6δ . On the full two-factor space put $h_R = h_0 + h_{e_z}$ and $P = P_0 P_{e_z} = |\Omega_R^0\rangle\langle\Omega_R^0|$, with lifted commuting projectors. The closed form inequality is

$$h_R \geq (I - P_0) + (I - P_{e_z}) = I - P + (I - P_0)(I - P_{e_z}) \geq I - P. \quad (141)$$

The four joint projection sectors prove the bounded-projector identity. Closure extends it from the finite tensor form core, without a representation cutoff or an assumption that the physical spaces tensorize.

AJ1's domain-valid mixed reset replaces R in a finite ground density by $P \otimes \text{Tr}_R \rho_\Lambda$. Its exterior marginal is unchanged, its R reference energy is zero, and its total form energy is finite. Bounded spectral truncations and monotone convergence justify the unbounded exterior energy cancellation. The variational principle cancels the finite ground scalar and all disjoint terms; each complete incident interaction changes expectation by at most $2M$. Therefore

$$\text{Tr}(\rho_{\Lambda,R} h_R) \leq 2M |\mathcal{I}_\Lambda(R)| \leq 4M = 28|\tau|. \quad (142)$$

This is an application of the inherited whole-group estimate, not a global small-norm perturbation argument. Pass each bounded spectral truncation of h_R through AJ1's actual local state limit, then increase the truncation. The actual normal density ρ_R obeys the same bound. By (141), its reference-vacuum deficit satisfies

$$e := 1 - \text{Tr}(\rho_R P) \leq \text{Tr}(\rho_R h_R) \leq 28|\tau| \leq \frac{7}{16384} =: e_*. \quad (143)$$

Multiplying the reference energy by δ converts its units; it does not make it the spectral first moment of the centered physical Wilson vector. At $\tau = 0$, zero deficit forces $\rho_R = P$. At nonzero admissible coupling it supplies an overlap bound, not purity.

11.2 A mixed-density trace estimate with its full normalization

For a unit vector ψ , the difference $|\psi\rangle\langle\psi| - P$ vanishes outside the span of ψ and Ω_R^0 . In a basis formed from the reference vacuum and its orthogonal component it has trace zero and eigenvalues

$\pm\sqrt{1 - |\langle\Omega_R^0, \psi\rangle|^2}$. The coincident and orthogonal endpoints satisfy the same formula. Thus its trace norm is twice that square root.

For an arbitrary positive trace-one density write its trace-class spectral decomposition $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$. Trace-norm convergence, the triangle inequality and weighted Cauchy–Schwarz give

$$\|\rho - P\|_1 \leq 2 \sum_j p_j \sqrt{1 - |\langle\Omega_R^0, \psi_j\rangle|^2} \quad (144)$$

$$\leq 2 \sqrt{\sum_j p_j (1 - |\langle\Omega_R^0, \psi_j\rangle|^2)} = 2\sqrt{1 - \text{Tr}(\rho P)}. \quad (145)$$

Partial sums and monotone convergence justify the countable case. This is an upper bound for mixed densities; imposing the pure-projector equality would be false. The symbol $T = \|\rho_R - P\|_1$ denotes the *full* trace norm, not its half-distance convention. Exact arithmetic in (143) gives

$$T \leq 2\sqrt{e}, \quad T^2 \leq \frac{7}{4096} < \frac{1}{576}, \quad T \leq \frac{1}{24}, \quad (146)$$

since $7 \cdot 576 = 4032 < 4096$. No floating square root or evaluated source constant enters this cap.

11.3 Actual reference moments, unknown mean and attributed refinement

The reference vector can contain entanglement within each selected strip, but every original z link is reference-free. Condition on all link variables except $u = U_z(0)$. The Wilson holonomy is Au^{-1} with A fixed in this conditional integral. Inversion and left translation preserve normalized Haar, and the remaining squared reference wavefunction is independent of u and integrates to one. For a uniform unit quaternion, sign symmetry gives zero mean of its scalar coordinate, and its four equal second moments sum to one. Consequently

$$\text{Tr}(PW) = 0, \quad \text{Tr}(PW^2) = \frac{1}{4}. \quad (147)$$

These are moments of the actual selected reference vector, uniform over its allowed coefficients. They are not an interacting conditional Haar law or an assumption of independent plaquette variables.

Let $m = \omega(W)$ be the actual unknown real mean. Trace duality and $\|W\| = \|W^2\| = 1$ give the common producer estimates

$$|m| \leq T, \quad \omega(W^2) \geq \frac{1}{4} - T, \quad v := \text{Var}_\omega(W) \geq \frac{1}{4} - T - T^2. \quad (148)$$

The polynomial $T + T^2$ is increasing on $T \geq 0$, since its difference at $y \geq x$ is $(y - x)(1 + y + x) \geq 0$. Thus

$$v \geq \frac{1}{4} - \frac{1}{24} - \frac{1}{576} = \frac{119}{576} = \frac{1}{5} + \frac{19}{2880}. \quad (149)$$

The squared mean is essential. Before this rational relaxation, the common route gives the lower expression $1/4 - 2\sqrt{28|\tau|} - 112|\tau|$.

The independent skeptic had already frozen a sharper second-moment argument before exchange. Since $\text{Tr}(\rho_R - P) = 0$ and $0 \leq W^2 \leq I$,

$$\left| \text{Tr}[(\rho_R - P)W^2] \right| = \left| \text{Tr}[(\rho_R - P)(W^2 - I/2)] \right| \leq \frac{T}{2}. \quad (150)$$

The signed first-moment estimate still costs T . Therefore this separately attributed proof gives

$$v \geq \frac{1}{4} - \frac{T}{2} - T^2 \geq \frac{131}{576} = \frac{1}{5} + \frac{79}{2880}. \quad (151)$$

The improvement is valid without another physical premise, but is not assigned retroactively to the two producers. Neither bound is sharpness or an actual variance measurement. At zero coupling the actual variance is exactly $1/4$.

11.4 Discriminating controls and admitted limits

The exact pure-projector fixtures reject removing the square root or the factor two from (145); mixed matrices reject imposing its pure-state equality on ρ_R . For example the reverse density $(\begin{smallmatrix} 17 & 6 \\ 6 & 8 \end{smallmatrix})/25$ has determinant $4/25$, deficit $8/25$ and trace distance $4/5$. Both defective upper bounds fail. The skeptic's signed-observable and positive-effect controls distinguish T from $T/2$; the effect saving cannot be copied to the mean of W .

On the same original configuration space, $\sqrt{1+W}\Omega_R^0$ is a normalized alternative normal vector. Conditional reference moments give mean $1/4$, second moment $1/4$ and variance $3/16 < 1/5$. It exposes a discarded squared mean or an unjustified interacting Haar substitution. Its nonzero mean and (145) require reference-vacuum deficit at least $1/64 > e_*$, so the added actual energy premise excludes it. It is not an interacting ground-state calculation.

Likewise, normalize $\mathbf{1}_{\{|W|<1/4\}}\Omega_R^0$ by the square root of its reference probability p . Conditional Haar gives $p > 0$, mean zero and positive variance at most $1/16$. The density $(2/\pi)\sqrt{1-w^2}$ and $\pi > 2$ give $p \leq 1/2$, hence deficit and reference energy at least $1/2$. These normal states violate (143); they show why normality alone gives no uniform floor. The possibly infinite energy of a discontinuous indicator is not claimed finite.

At the origin, counting only outgoing anchors or only individually touching faces is blind: all incident anchors lie in R and all their group faces touch it. The retained incidence-only region $R' = \{(1, 1, 1), (1, 1, 2)\}$ resolves this. Its seven orthant anchors yield four and seven retained groups in cubes of sides three and four, versus one and two outgoing anchors. The inherited group budget charges 84 and 147 faces, versus 49 and 82 individually touching faces. These diagnostics reject substituting a different count into AJ1's whole-group estimate. They do not rule out a separately derived estimate for a finer decomposition into face interactions, and establish no translated-observable variance theorem.

Scalar and scale controls retain $E_{\text{strip},b}$ and $\delta = \alpha/8$. Full noncommuting original-link gauge transformations preserve W while missing-head actions fail. Hypothetical positive source constants with $\tau_* < 2^{-16}$ reject treating the cap as an evaluated stability radius; those controls do not assign the actual source constants.

The advisor suggested the overlap route before production. Separate forward, reverse and independent skeptical derivations froze before exchange; the stronger effect estimate was in that independent freeze. This skeptic's new AK1 derivation is distinct from its earlier AJ2 reviewer-handoff closeout. The post-review read both full reports and checkers and replayed both complete output sets under normal and optimized Python with byte equality. Administrative input-path and inventory-selector histories, blind incidence controls and first successful scientific outputs remain preserved. They add no research loop or independent observation.

The controlling AJ1/AJ2 proofs were read by all three workers. Forward and skeptic read the complete current-manuscript text; reverse recorded focused reading. The skeptic's text reading is not visual PDF review. The I1 stability theorem remains inherited, and no new external primary retrieval or scientific-priority comparison is claimed. Finite rational and geometry diagnostics audit this proof without replacing its analytic infinite-volume premises.

AK1 proves the variance floors for the fixed original observable and specified conditional homogeneous state. It proves no physical excited-vector form or operator domain, first energy moment, commutator sum rule, finite spectral window, lower heat bound, numerical mass or AK2 theorem. Other translates, boundary states and models require their own estimates. No physical matching, continuum Yang–Mills construction, historical validation or defensible percentage of the Millennium problem follows.

12 AK2: physical Wilson form energy, a finite window and lower heat^{**}

For the same actual homogeneous state and original origin xz Wilson observable, AK2 proves

$$\chi \in D(H_{\text{phys}}^{1/2}), \quad \mu_1 := \int E d\nu(E) \leq \alpha, \quad \nu([\alpha/16, 10\alpha]) \geq \frac{1}{10}, \quad (152)$$

and, for every finite physical imaginary time,

$$C(t) = \langle \chi, e^{-tH_{\text{phys}}/\hbar} \chi \rangle \geq \frac{1}{5} e^{-5\alpha t/\hbar}, \quad 0 \leq t < \infty. \quad (153)$$

Here $\chi = (\pi(W) - \omega(W)I)\Omega$ and ν is its actual physical energy spectral measure. The regime remains *both* $|\tau| < \tau_*$ with unevaluated source constants and $|\tau| \leq 2^{-16}$, with the same fixed selected coefficients, positive scales and whole-star positive-orthant limit. The numerical cap alone certifies no chosen positive admissible coupling.

The proof uses only the common AK1 consequence $v = \|\chi\|^2 \geq 1/5$. Its producer floor 119/576 and independent-skeptic refinement 131/576 retain their attribution and are not needed to optimize this target. AJ1/AJ2 supply

$$\nu(\mathbb{R}) = v, \quad \nu(\{0\}) = 0, \quad \text{supp } \nu \subset [\alpha/16, \infty), \quad C(t) \leq v e^{-\alpha t/(16\hbar)}. \quad (154)$$

AK1's local reference-density energy is not a physical first moment of χ . The new calculation supplies that missing premise from the actual original-link kinetic operator. It proves a form-domain statement and a first-moment upper bound, without asserting $\chi \in D(H_{\text{phys}})$, a second moment, a threshold eigenatom or a measured mass.

12.1 Finite physical operator, complete support and multiplier domains

Retain the full $R = \{0, e_z\}$, all 48 original links and all 36 endpoint actions from AJ2/AK1. The four distinct signed Wilson links have owners $0, 0, e_z, 0$; all other 44 local links remain present even though their derivatives of W vanish. Each finite volume contains every owned link, three selected faces per block and every complete retained 21-face star. For a coarse cube of side n , the original counts are $24n^3$ links, $8n^3 + 14n^2$ endpoints, $3n^3$ selected faces and $21(n-1)^3$ retained omitted faces. The original tail ownership and complete-star rule establish these formulas at every volume; the $n = 2, 3, 4$ diagnostics are finite checks of that geometry.

On the finite product Haar space, choose the standard half-Pauli left fields

$$X_{e,a} f(U_e) = \left. \frac{d}{dt} f(e^{tT_a} U_e) \right|_{t=0}, \quad T_a = -i\sigma_a/2, \quad C_e = - \sum_{a=1}^3 X_{e,a}^2. \quad (155)$$

Thus a fundamental matrix coefficient has Casimir eigenvalue $3/4$. Reversing the overall Lie-algebra basis convention, as in the forward derivation, leaves this Casimir and the squared-gradient identity unchanged; the signs within each inverse-link derivative still matter.

Write the raw physical operator and reference scalar as

$$H_{\Lambda}^{\text{raw}} = \alpha \sum_{e \text{ owned}} C_e - \sum_{f \text{ selected}} \lambda_f W_f - \frac{\alpha\tau}{24} \sum_{b+S \subset \Lambda} \sum_{f \in O_b} W_f, \quad (156)$$

$$B_{\Lambda} = \sum_{b \in \Lambda} E_{\text{strip},b}, \quad \delta \hat{H}_{\Lambda} = H_{\Lambda}^{\text{raw}} - B_{\Lambda}, \quad \delta = \alpha/8, \quad |O_b| = 21. \quad (157)$$

The selected λ_f include both endpoint and bridge coefficients. If $\hat{E}_{\Lambda}$ is the actual normalized ground energy, then

$$E_{\Lambda}^{\text{raw}} = B_{\Lambda} + \delta \hat{E}_{\Lambda}, \quad K_{\Lambda} = \delta(\hat{H}_{\Lambda} - \hat{E}_{\Lambda}) = H_{\Lambda}^{\text{raw}} - E_{\Lambda}^{\text{raw}} \geq 0. \quad (158)$$

Both reference and true ground scalars are retained before cancellation. For the actual normalized ground Ω_Λ , put

$$m_\Lambda = \langle \Omega_\Lambda, W \Omega_\Lambda \rangle, \quad \chi_\Lambda = (W - m_\Lambda) \Omega_\Lambda, \quad K_\Lambda \Omega_\Lambda = 0. \quad (159)$$

The finite sum of link Casimirs is an elliptic product Laplacian with operator domain H^2 and form domain H^1 on the compact group product. Finite Peter–Weyl sums form a graph core. Every selected and omitted potential is smooth real bounded multiplication at each fixed volume, so bounded perturbation preserves these domains. A uniform bound on the entire extensive potential is not assumed. In particular the finite ground is in H^2 .

Smoothness, rather than bounded locality alone, supplies the multiplier regularity. Both W and W^2 have bounded derivatives through order two. On the finite core, for $F = W$ or W^2 and $Q_\Lambda = \sum_e C_e$,

$$Q_\Lambda(F\psi) = FQ_\Lambda\psi + (Q_\Lambda F)\psi - 2 \sum_{e,a} (X_{e,a}F)X_{e,a}\psi. \quad (160)$$

The first-derivative norm satisfies $\sum_{e,a} \|X_{e,a}\psi\|^2 = \langle \psi, Q_\Lambda \psi \rangle$ and is bounded by the graph norm using $x \leq 1 + x^2$. A graph-approximating sequence and closedness therefore prove $FD(Q_\Lambda) \subset D(Q_\Lambda)$. The weak product rule similarly gives H^1 preservation. The same statements hold for $W - m_\Lambda$ and its square. These are the required finite-volume domains; no infinite-volume operator-domain conclusion is imported from them.

12.2 Four original derivatives and the exact finite form identity

On the common finite core, Haar integration by parts and the product rule give

$$[C_e, W]\psi = (C_e W)\psi - 2 \sum_a (X_{e,a} W)X_{e,a}\psi, \quad [W, [C_e, W]] = 2M_{\sum_a (X_{e,a} W)^2}. \quad (161)$$

The first commutator is a first-order operator, not an asserted bounded L^2 operator. The second has the displayed bounded multiplication extension; its domain identity extends to H^2 by the preceding argument. All selected and retained omitted magnetic terms commute with W as multiplication operators, including overlapping plaquettes. Both scalars commute as well. Their cancellation does not replace the actual interacting ground by Haar.

For a positive original link occurrence, varying U by $e^{tT_a}U$ differentiates it as $T_a U$. For an inverse occurrence,

$$(e^{tT_a}U)^{-1} = U^{-1}e^{-tT_a}, \quad \left. \frac{d}{dt}(e^{tT_a}U)^{-1} \right|_0 = -U^{-1}T_a. \quad (162)$$

Both occurrences have second derivative $-U^{\pm 1}/4$. Fixed left/right factors and inversion are isometries of the bi-invariant metric. If the full holonomy is the unit quaternion $(w, \mathbf{q})$, then $W = w$ and $|\mathbf{q}|^2 = 1 - w^2$. The three half-Pauli fields give scalar-coordinate derivatives equal to components of $\mathbf{q}/2$, up to an orthogonal adjoint rotation and the required inverse sign. Thus *each* of the four distinct original links gives

$$\sum_a (X_{e,a} W)^2 = \frac{1 - W^2}{4}, \quad C_e W = \frac{3}{4}W. \quad (163)$$

Every other link derivative is zero. Retaining all four original kinetic terms yields

$$\Gamma(W) := \sum_{e,a} (X_{e,a} W)^2 = 1 - W^2, \quad [W, [K_\Lambda, W]] = 2\alpha M_{1-W^2}. \quad (164)$$

The identity does not substitute a reduced one-plaquette kinetic operator.

The previously proved domains make $WK_\Lambda W$, W^2K_Λ and $K_\Lambda W^2$ well-defined at the ground vector. In $2WK_\Lambda W - W^2K_\Lambda - K_\Lambda W^2$, the last two ground expectations vanish. Centering W leaves its form energy unchanged because $K_\Lambda \Omega_\Lambda = 0$. Therefore, for the actual finite spectral measure ν_Λ ,

$$\mu_{1,\Lambda} = \langle \chi_\Lambda, K_\Lambda \chi_\Lambda \rangle = \frac{1}{2} \langle \Omega_\Lambda, [W, [K_\Lambda, W]] \Omega_\Lambda \rangle \quad (165)$$

$$= \alpha \omega_\Lambda (1 - W^2) \leq \alpha. \quad (166)$$

The forward proof independently obtains this by subtracting the weak ground equation tested against $f^2 \Omega_\Lambda$ from the form at $f \Omega_\Lambda$, $f = W - m_\Lambda$. For each real derivative,

$$|X(f \Omega_\Lambda)|^2 - \operatorname{Re}(\overline{X \Omega_\Lambda} X(f^2 \Omega_\Lambda)) = (Xf)^2 |\Omega_\Lambda|^2. \quad (167)$$

The potential and scalar terms cancel, and H^1 continuity extends the identity from the smooth core. Both routes give the same physical coefficient and factor one-half. The bound is uniform in volume and the declared coefficients, despite their effect on the unevaluated actual expectation.

12.3 Tested resolvents, scalar measure convergence and the limiting form

The source-qualified I1/AJ1 theorem gives centered finite-resolvent matrix-element convergence tested on fixed bounded-local vectors $A\Omega_\Lambda, B\Omega_\Lambda$. Its domain-qualified creation core is retained separately. The theorem does not assert common-concrete-space strong resolvent convergence or place every bounded-local vector in $D(G)$. Fixed physical energy is restored through

$$(K_\Lambda - z)^{-1} = \delta^{-1}(\hat{H}_\Lambda - \hat{E}_\Lambda - z/\delta)^{-1}, \quad \operatorname{Im} z \neq 0. \quad (168)$$

Local state convergence gives $m_\Lambda \rightarrow m = \omega(W)$ and $v_\Lambda = \omega_\Lambda(W^2) - m_\Lambda^2 \rightarrow v$, with $v_\Lambda \leq 1$. First use the source theorem on the fixed $A = W - mI$. Replacing its finite cyclic vector by χ_Λ changes it by $(m_\Lambda - m)\Omega_\Lambda$. The resolvent norm $1/|\operatorname{Im} z|$ and both vector norms at most two bound the difference of quadratic resolvent tests by

$$\frac{4|m_\Lambda - m|}{|\operatorname{Im} z|} \rightarrow 0. \quad (169)$$

This justifies the moving ground-state mean. The resulting nonreal transforms converge to those of the actual spectral measure ν of (H_{phys}, χ) ; AJ1's reducing physical identification preserves this measure.

The linear span of nonreal resolvent functions is dense in $C_0(\mathbb{R})$. Conjugation stays in the span, products at distinct poles reduce by the resolvent identity, and repeated poles are uniform limits of difference quotients with poles away from the real line. Its uniform closure is therefore an algebra that separates points and vanishes nowhere. The locally compact Stone–Weierstrass theorem gives the claimed density. The uniform measure-mass bound extends convergence to every C_0 test. The reverse proof equivalently works on $[0, \infty)$.

For $L > 0$ and $E \geq 0$ use the continuous compactly supported functions

$$F_L(E) = E \min\{1, \max(0, 2 - E/L)\}, \quad F_L(E) = 0 \quad (E < 0). \quad (170)$$

They satisfy $0 \leq F_L \leq E$ on the positive half-line and increase pointwise to E . The measure convergence and (166) give $\int F_L d\nu \leq \alpha$ for every L , and monotone convergence yields

$$\mu_1 = \int E d\nu(E) \leq \alpha, \quad \left\| H_{\text{phys}}^{1/2} \chi \right\|^2 = \mu_1 < \infty. \quad (171)$$

This is the spectral characterization of the actual form domain. The independent skeptic instead uses $\min(E, L) = L - (L - |E|)_+$ on nonnegative energies: mass convergence and the C_0 test give the same monotone argument.

Equation (166) is an *equality at finite volume*. Its bounded-local right side converges, but the limiting spectral first moment is only bounded above by this method. Uniform integrability has not been established. Neither equality of first moments, a second moment nor $\chi \in D(H_{\text{phys}})$ is inferred. Scalar measure convergence replaces no missing common-Hilbert-space operator theorem.

The independently frozen skeptic derivation also retains the more specific upper bound $\mu_1 \leq \alpha[1 - \omega(W^2)]$, by passing the bounded-local right side through the same argument. It remains an inequality with an unevaluated interacting expectation. The common ceiling α suffices for the frozen window and heat targets below.

12.4 A bounded energy interval and all finite imaginary times

With $\text{supp } \nu \subset [\alpha/16, \infty)$, $v \geq 1/5$ and (171), the elementary tail inequality gives

$$\nu((10\alpha, \infty)) \leq \frac{\mu_1}{10\alpha} \leq \frac{1}{10}, \quad \nu([\alpha/16, 10\alpha]) \geq v - \frac{1}{10} \geq \frac{1}{10}. \quad (172)$$

The removed tail is open and the window's upper endpoint is included. Positive interval mass does not identify an eigenatom, the lowest overlapping energy or a sharp threshold.

Normalize the positive measure by v to obtain a probability measure with finite mean $\bar{E} = \mu_1/v \leq 5\alpha$. For $s = t/\hbar \geq 0$, the convex tangent inequality

$$e^{-sE} \geq e^{-s\bar{E}}[1 - s(E - \bar{E})] \quad (173)$$

is integrable because the first moment is finite. Integrating proves Jensen's inequality here without a small-time expansion:

$$C(t) \geq ve^{-t\mu_1/(\hbar v)} \geq ve^{-5\alpha t/\hbar} \geq \frac{1}{5}e^{-5\alpha t/\hbar}. \quad (174)$$

It includes $t = 0$, where $C(0) = v$. Together with (154), this gives two physical imaginary-time envelopes for every finite time. The variance, spectral mass and $C(t)$ are dimensionless; α , μ_1 and the window endpoints have energy units, and $\alpha t/\hbar$ is dimensionless. No real-time magnitude-decay or exact asymptotic mass is claimed.

12.5 Controls, preserved failures and the stopping boundary

At the explicitly labeled corner where *all* selected and omitted couplings vanish, the Haar vacuum gives $\|W\Omega\|^2 = 1/4$, excitation energy 3α and first moment $3\alpha/4$. This checks the four-link normalization, without assigning those values to an arbitrary interacting state. Doubling the half-Pauli generator quadruples the gradient and Casimir budget. Omitting the double-commutator factor one-half gives the wrong moment. Scalar and units fixtures separately detect a missing actual ground subtraction, δ conversion or division by $\hbar$; simultaneous raw-operator and ground shifts cancel.

Exact noncommuting quaternion derivatives, supplemented by signed group-displacement or Cayley-path tests, verify all four original occurrences and their inverse derivatives. Squaring an inverse derivative hides its wrong overall sign, so that blind gradient test is preserved beside signed first-derivative controls. The independent skeptic's first identity-link fixture has $\Gamma = 0$ and is also blind to a kinetic rescaling; the later noncommuting fixture discriminates. Full endpoint gauge transformations preserve the actual word, while missing-head actions fail.

Both producers give abstract bounded self-adjoint creating operators with vectors of divergent form energy. The skeptic additionally uses actual free original-link characters, with an auxiliary observable distinct from W . On the full 48-link region, let ϕ_n be the spin- $n/2$ character of the same

four-link holonomy and let Ω_R^0 be its free constant vacuum. Conditional Haar gives orthonormal ϕ_n , with four-link energy $\alpha n(n+2)$. The gauge-invariant vector

$$f_R = \sqrt{3} \sum_{k \geq 1} 2^{-k} \phi_{2^k} \quad (175)$$

has norm one and is orthogonal to Ω_R^0 . Its first N terms have squared norm $1 - 4^{-N}$ and form energy divided by α equal to $3N + 6(1 - 2^{-N})$, which diverges. The bounded local physical self-adjoint operator

$$A_R = |f_R\rangle\langle\Omega_R^0| + |\Omega_R^0\rangle\langle f_R| \quad \text{on } \mathcal{K}_R, \quad A = A_R \otimes I_{R^c} \quad (176)$$

has norm one and creates the divergent-energy vector in the free product representation. It is extended by the exterior *identity*, not a global vacuum projector. The example rejects automatic form regularity of bounded local observables; it does not undermine the separately proved smooth-Wilson argument.

The three independent moment-loss diagnostics are abstract positive spectral measures. For example the reverse uses, in units of α ,

$$\eta_n = (1 - 1/n)\delta_{1/16} + (1/n)\delta_{1/16+n/2}. \quad (177)$$

Masses are one, bounded-test differences from $\delta_{1/16}$ are at most $2\|f\|_\infty/n$, and every first moment is $9/16$, while the limiting first moment is $1/16$. This proves that bounded first moments and mass/resolvent convergence do not force moment equality. The forward and skeptic retain their different valid measures and exact prefixes.

Noncentered observables with nonzero means retain vacuum mass and heat plateaus. By contrast, abstract mass $1/5$ at 20α has the inherited lower support and variance floor but neither mass in the target window nor the proposed lower heat curve; its first moment exceeds α . The skeptic uses 11α for the same inference control and separately checks inclusion of 10α in the closed window. These are alternative measures, not observed spectral atoms of the actual state. Hypothetical source constants below the numerical cap retain the strict symbolic coupling requirement.

The forward first invocation stopped at a missing closing parenthesis before executing science. The original checker and SyntaxError are preserved, followed by corrected normal and optimized output agreement. Blind controls and replacements remain attributed; none is erased or counted as another investigation. Frozen sources and independent implementations support reproducibility, while the finite domain proof, original-link algebra, source-qualified measure limit and monotone convergence establish the result. The original I1 cluster-expansion theorem remains inherited; no full reproof, new primary retrieval or scientific-priority claim is made.

The final post-review also preserves one unused stale label in a reverse centering diagnostic: its metadata says `abstract_gap=1`, while the implemented positive level is 4 in units of α . The correctly implemented centered mass is $1/16$, first moment divided by α is $1/4$, and centered and raw heat values are $1/256$ and $65/256$. The reverse author's Newton-method comparison acknowledges the stale label; it is not reinterpreted as an intentional lower-bound certificate. No accepted target depends on it, and no frozen field is rewritten. That comparison shares authorship with reverse and is not another independent producer.

The final skeptic authored its AK2 derivation before exchange, then read both complete producer reports and checkers and reproduced each complete output set under normal and optimized Python with byte equality. Its separately bound final specification audit found no blocking issue. The local character diagnostic in (176) is the additive post-review clarification of its own independently frozen control. Source reading remains differentiated: the skeptic's current manuscript coverage was differential text coverage against its previously read draft, while both producers recorded focused current readings. These activities are not visual PDF review or fresh audits of the full external cluster proof.

This is investigation ten and the authorized scientific stop boundary. The fixed-lattice, fixed-observable result supplies (152)–(153) in the same conditional homogeneous model. It supplies no numerical stability radius, infinite first-moment equality, second moment, infinite operator-domain assertion, eigenatom, exact mass, other-boundary identification, physical model matching or continuum Yang–Mills theorem. Further questions remain plans rather than extra scientific executions.

13 Completed investigation count and remaining obligations

All ten requested investigations are admitted, completing five adaptive goal pairs. This is 100% of the requested investigation count, including limited outcomes. It is not a percentage of the Yang–Mills existence-and-mass-gap proof: equations, loops, numerical error percentages and network size provide no calibrated denominator for that theorem.

The final assessment identifies the strongest structural advance as a controlled, nonvacuous physical sector of the declared fixed-spacing $SU(2)$ homogeneous omitted interaction on its selected-strip reference. AJ1 supplies the normal local representation and reducing physical generator; AJ2 supplies a nonzero original Wilson fluctuation; AK1/AK2 add conditional variance, physical form-energy, finite-window mass and imaginary-time control. These are improvements within the specified whole-star orthant state. The inherited $\alpha/16$ threshold is not a new continuum gap, and the unnormalized Wilson spectral mass is not a particle count or an evaluated mass.

The final retrospective assessment is one contemporary model-agent review applying the documented historical-method questions. It is not five new independent reviewers, participation by historical figures, external human peer review or another scientific investigation. The independent productions share inherited premises and model authorship. The Newton AK2 comparison is by the reverse producer; the separate skeptical production and later clarifications retain their recorded attribution.

The advisor’s 22 September 2026 [official problem-status check](#) records the problem as unsolved. Selected sections 3–5 and 6.5 of the [Jaffe–Witten description](#) were checked for scope, not as an exhaustive current research survey. The required nontrivial four-dimensional theory for any compact simple gauge group, suitable axioms, required short-distance behavior and strictly positive finite physical mass gap remain open here. Confinement and an isolated one-particle state are discussed extensions, not additional mandatory clauses of the stated Millennium problem. Numerical lattice stability constants, a particular correction iteration and exact Wilson moment equality are possible technical milestones, likewise not compulsory clauses in themselves.

For this workbench’s proposed route, full generator, observable, state, coupling, spacing and clock matching remains unresolved. So does identification of specified bulk and boundary limits. A boundary-local orthant witness cannot silently become a bulk observable, and the finite, canonical, summable and homogeneous models cannot be interchanged. The finite first-moment equality has not been proved to survive the limit; no actual strict moment loss is established.

The final roadmap ranks the following *unselected* questions. Its ordering concerns missing implications and bounded test value, not success probabilities or equal units of proof work.

Goal	Proposed direction	First unresolved premise
AL	Full model and physical-scale bridge	A complete coefficient, observable and clock dictionary with one admissible scaling path; compare the existing bare-coupling obstruction before claiming a new bridge.
AM	Numerical homogeneous stability interval	Evaluate every constant required for the complete original model, retaining an insufficient verdict if a necessary constant or infinite estimate remains open.
AN	Specified bulk-boundary comparison	Compare deep-bulk orthant translates with one declared centered-box route using a bound that vanishes with boundary distance.
AO	Same-Wilson moment identification	Prove sufficient uniform integrability or form convergence before transferring the finite identity as equality; assume no new energy-tail or second-moment theorem.
AP	Canonical inference at a prior-selected design	Separately test one design fixed from coarse prior information and decisions across independently varying candidate parameters.

Every second loop remains unselected until its first review. A certified numerical evaluator for the declared 561-state AH enrichment is a lower-priority computational alternative; no such

heat vector was produced by this cycle's assessment. The detailed models, dependency IDs and proposed first tests are recorded in `research/round28/advisor/roadmap.json`. Investigation ten is the stopping boundary. This closeout executes no further research target.

The assessment, roadmap and official scope check are separately source-bound. They do not alter the 33-record exploratory reading ledger or claim new primary research by the retrospective reviewer. Scientific priority remains unverified; the two-star marker identifies contributions in this workbench.

A Admitted scope and contribution ledger

Loop	Admitted contribution	Possible application	Remaining limitation
AG2	Complete original remainder and transport; output weight-two budget below 0.007, and centered-reference gap above 0.995 on the main interval.	Inventory the missing original interaction and supply explicit scalar, mixing and reference-diagonal budgets for the next review.	The remaining mixing is retained. No gap of the full corrected Hamiltonian, complete-source contraction or next iteration is proved.
AG3	Complete one-step correction of actual AG2 mixing, with coefficients 3060/3623 and 5949/96812 from weight 2 to 3/2; full scalar and reference-diagonal updates.	Control one complete original-source correction and compare it with passive reweighting before choosing a further method.	Main coefficient is weaker than passive reweighting. Narrow coefficient beats only its universal upper ceiling. No actual lower-weight improvement, same-weight contraction, iteration or full gap.
AI3	Exact odd moments and an all-order retained reflection-tail bound. All ten complete ratio certificates remain insufficient; both direct scalar probes still separate at $u=1e-18$, $k=5$.	Remove the material cubic arithmetic floor and identify the state or spatial upper budget that a later physical readout method must improve.	Finite-grid gain comes primarily from degree-eight centers. No physical error cancellation, continuous inverse, unknown-phase tolerance, homogeneous matching or mass-gap result.
AI4	Complete physical ratio intervals separate by at least $39u/40$ for every real $0 < u \leq 1e-24$ under the dyadic collar and shorter-duration schedule.	A uniform family of conditional candidate-pair tests with proved all-depth resource envelopes and attributed scalar-error allowances.	Clock grows as the inverse cube of u ; all scalar allowances vanish with u . No fixed-error apparatus, continuous inverse, automatic enlarged-graph theorem, homogeneous matching or continuum result.
AH1	Complete new 46-link graph; strict rank-30 cutoff and rank-561 physical enrichment, exact Gram/magnetic action, and all-time relative heat error below 0.0022 for the declared preparation ball.	A complete finite-graph generator and conservative heat certificate that retains every new-input outside channel and both ground comparisons.	Exact own-ground heat only; no numerical trajectory, whole-shell completion, evaluated outside Gram, real-time theorem, graph-size uniformity or continuum result.
AH2	Complete 531-channel delayed loading with feedback gives a proved all-time true-relative exact heat error below 0.0001 on the unchanged graph and preparation ball.	Improve the same-clock physical omission certificate by integrating stationary and excited loading before applying the full outside bound.	Sufficient bound for exact separate ground centers; no numerical heat vector, evaluated outside Gram, real-time theorem, graph-size uniformity, model matching or continuum result.

Loop	Admitted contribution	Possible application	Remaining limitation
AJ1	Actual homogeneous I1 local energy bound, local trace-norm limits and normal representations; compatible Haar averaging, physical cyclic/fixed equality and closed reducing generator.	Identify the physical GNS restriction of the specified homogeneous orthant state with justified gauge continuity, local averages and operator/form domains.	Symbolic coupling interval and inherited $\alpha/16$ threshold only; no nonzero physical excitation or Wilson variance, numerical interval, boundary equivalence, model matching or continuum result.
AJ2	The actual fixed xz Wilson observable has positive variance in the homogeneous state; its nonzero physical spectral measure gives strictly positive finite-time heat correlation and the inherited $\alpha/16$ upper decay ceiling.	Close the physical-sector nonvacuity omission using Haar-null multiplier levels, local normality, the physical GNS quotient and bounded spectral calculus.	Symbolic coupling interval only; no quantitative variance floor, energy moment, threshold atom, measured mass, lower decay law, real-time magnitude decay, boundary identification or continuum result.
AK1	Actual origin Wilson variance at least $119/576$ from both producers; the independent skeptic improves the lower certificate to $131/576$ on the same conditional subregime.	Turn actual local reference energy into a mixed-state vacuum-overlap bound and a quantitative physical signal margin, retaining the unknown squared mean.	Requires both symbolic stability and the additional coupling cap $1/65536$; no evaluated stability radius, actual variance, physical excited-vector energy moment/domain, spectral window, translated theorem or continuum result.
AK2	Actual Wilson physical form domain and first moment at most α ; spectral mass at least $1/10$ in $[\alpha/16, 10\alpha]$, and lower heat envelope $(1/5) \exp[-5\alpha t/\hbar]$ at every finite nonnegative time.	Transfer the finite original-link energy identity through tested resolvents and nonnegative cutoffs, then combine the upper moment with the admitted variance and support.	Same conditional fixed-lattice regime only; no limiting moment equality, infinite operator domain, second moment, spectral atom, evaluated mass, real-time decay, model transfer or continuum result.

B Additive source deepening after the first two goal pairs

The initial survey remains **30 formal reading records at 38 distinct URLs**. The later source-only supplement adds three attributed records: further selected passages of the already known Del Vecchio–Fröhlich–Pizzo paper, an ETH institutional publication record, and a targeted topology reading of the already known Yarotsky paper. These are not three new independent studies. Together the current ledgers contain **33 formal reading records at 41 distinct recorded URLs**, plus five screening or reopen records; the URL union including those screens is 46. The intermediate two-record supplement had totals 32/39 and a screened URL union of 44; its creation checkpoint remains preserved. The one unread lead stays outside the reviewed count. No government-primary document was newly read in this supplement, and it adds zero research investigations.

The technical text is [arXiv:2108.13907v1](#), a 54-page preprint. The [ETH institutional record](#) verifies publication in *Journal of Functional Analysis* 284(1), 109734 (2023), [doi:10.1016/j.jfa.2022.109734](#). Final journal text was not compared with v1. The recorded technical passages are Sections 1.1–1.2 and Remark 1.1; the Section 2 opening; definitions (2.29)–(2.32); Lemma 2.8’s statement and initial proof; selected Section 2.4 gap estimates through Corollary 2.12; the Section 3 induction introduction and selected Theorem 3.1 inequalities; Theorem 3.3 and its closing proof; Lemma 4.1’s statement; and Lemma 4.3 with the adjoining inverse-series justification. The long induction and appendices were not audited end to end.

The Feynman-method deepening initially compared an application to the original finite-range homogeneous Hamiltonian $H_0 + \Phi$ with organizing AG3’s generated interaction inside the cited paper’s evolving-reference induction. Its later archive correction identifies an omitted established result: the current Round21 I1 admission, revision 2, already supplies the complete 24-link/21-face dictionary, qualitative original-model stability and the specified positive-orthant full-link-observable GNS construction. With source constants left symbolic,

$$\tau_* = \frac{\min\{c_1(S), 1/[2c_2(S)]\}}{7} > 0, \quad 0 < |\tau| < \tau_* \implies \Delta \geq \alpha/16. \quad (178)$$

I1 certifies no selected nonzero numerical τ . Its homogeneous *omitted* coupling on the selected-strip reference is not a uniform coefficient on every plaquette, a summable-model representation, or a continuum construction. The specified orthant GNS conclusion is not independence of every boundary sequence.

Consequently another theorem application giving only unspecified-small-coupling existence would duplicate the admitted qualitative outcome. A distinct new target must supply a numerical interval, complete generated-interaction iteration, or an additional representation or boundary statement beyond I1. The archive comparison uses the current repository admission and source dictionary; it neither rereads nor reconstructs the external Yarotsky proof, and adds no primary reading record or investigation. The unchanged deeper-source note and this additive correction are separately bound in `research/round28/experts/source-supplement.json`.

A different published theorem would need its own hypothesis matching while reusing I1’s established site, star, domain and normalization dictionary. Infinite onsite dimension is permitted by the Del Vecchio paper’s stated assumptions; its initial range condition is not confined to the expository nearest-neighbor presentation. Theorem-specific boundary and gauge-observable restrictions must remain explicit, and a constrained physical space cannot silently become a tensor product. Re-entry with AG3’s generated interaction additionally requires a proved passage from arbitrary indexed supports to rectangular energy-weighted data, changed local gaps, retained scalars and diagonals, and the complete induction. These are inferred transfer obligations, not defects asserted in either external theorem or missing work already discharged by I1.

The third supplemental reading, record R28-FGNS1, inspects Yarotsky’s 17-page [math-ph/0411042v1 PDF](#) and corresponding [HTML](#). Its depth is the definitions and Theorems 1–3 on

printed pages 2–4, creation operators on pages 6–7, selected expansion and state-limit passages on pages 8–9, and the ending generator/core argument on pages 9–10. References on page 17 were bibliographic pointers only. The long proof and cited predecessor papers were not independently reconstructed; an older I1 download hash is not claimed as a hash of these newly inspected browser bytes.

The source states convergence of finite-ground expectations for each observable in the full local algebra $\bigcup_F B(\mathcal{H}_F)$, and resolvent matrix-element convergence tested on local excitation vectors. Its generator discussion uses local bounded commutators and creation vectors with $u_I \in \mathcal{H}'_I \cap D(H_{I,0})$, followed by density and essential self-adjointness statements. These are not trace-norm convergence of local density matrices, strong resolvent convergence in the earlier summable representation, or a declaration that every bounded-local cyclic vector lies in the generator domain. No explicit local-normality or gauge-average compatibility claim was located in the inspected passages; this limited reading does not prove that such properties fail or cannot follow from the full source theory.

For a later homogeneous physical-GNS identification, the source note therefore distinguishes pointwise state limits and a generator core from the regularity needed to pass local weak-operator Haar averages through the actual representation. It calls for local normality or another explicit continuity-and-averaging argument, continuous gauge implementers on all endpoint groups, the full bounded local invariant algebra, and a justified reducing spectral restriction. I1 already supplies the specified homogeneous full-algebra state and generator; J2's cyclic/invariant identification belongs to its dyadic summable representation and does not automatically transfer. These are prospective transfer obligations only. The reading constructs no vacuum-reset bound, local tightness estimate, gauge-cyclic equality or new gap.

The AG3 passive comparison and spent $9/4 \rightarrow 2 \rightarrow 3/2$ margins remain unchanged. AG2/AG3's complete-remainder and one-step results do not evaluate I1's source constants; their numerical extracted-reference gaps are not new numerical full-Hamiltonian gap theorems. The supplement supplies no new estimate, fixture, threshold, theorem application or continuum premise. It informed later planning without selecting a fourth or fifth goal at the time of the review. The source and reading-depth records, including the correction, are included in the manuscript source manifest.

The later admitted AJ1 investigation, Section 9, now supplies the previously prospective local-energy and normality argument, compatible finite-endpoint averaging, physical cyclic/fixed equality and closed reducing restriction for the actual I1 state. Its post-review clarification uses local normality and finite spectator columns to justify represented product-vector density; it does not assume operator-norm density of the algebraic tensor product. This mathematical admission is distinct from the earlier source-only note. AJ1's primary retrieval records and restricted proof-reading claims are bound through its gate; they do not silently change the 33-record survey ledger. AJ2, Section 10, then proves a nonzero homogeneous Wilson fluctuation using this normality and the actual multiplier's Haar-null levels. The coupling interval remains symbolic, and neither investigation evaluates a mass or the source stability constants.

C Reproduction and artifact integrity

The frozen contracts, producer reports, exact outputs and skeptical gates in `research/round28/` govern each result. Scientific replay is separate from building this manuscript. From the repository root the addendum is built with

```
python3 -B papers/round28-addendum/build.py
```

All ten investigations have source-bound advisor gates and a reviewed chapter here. The final build requires those ten gates and verifies their complete bindings. The explicitly marked intermediate-document mode `--draft` remains available; it includes only admitted chapters and writes solely to its temporary build directory. Neither mode promotes a report to acceptance. Figures are computed from admitted exact evidence; floating-point conversion is only for rendering unless a figure explicitly states a different numerical scope.

The build uses ordinary L^AT_EX, Poppler and the installed Python scientific stack. It installs no dependencies. Final publication requires rendering and visually reviewing every page after the last meaningful change. A clean compiler log, text-extraction check or targeted intermediate review does not replace that all-page inspection. The final artifact manifest binds the editable sources, figures, PDF and review record; a separate source manifest binds the admitted reports, outputs and gates. Hash agreement establishes integrity, not mathematical truth or independent experimental confirmation. Draft01 and the Round27 artifact bytes remain unchanged.

Open target. The admitted homogeneous representation and Wilson results, canonical readout certificates, finite-graph calculations and reference-only bounds retain their separate models and conclusions. Their combination does not supply the missing continuum construction, axioms or physical matching. The scope ledger states each investigation’s result, possible application and remaining limitation; later results do not silently rewrite earlier admissions.